Are Lengthy and Boilerplate Risk Factor Disclosures Inadequate? An Examination of Judicial and Regulatory Assessments of Risk Factor Language

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ABSTRACT

Although formal guidance instructs firms to avoid issuing lengthy and boilerplate risk factor disclosures, regulators and users of financial statements note these disclosures continue to be excessively long and boilerplate. The persistence of these characteristics is particularly surprising given that prior research finds firms disclosing lengthy and boilerplate risk factors experience negative capital market consequences. We investigate two potential sources of firms' incentives to issue such disclosures by examining how judicial and regulatory assessments of firms' risk factor disclosures correlate with measures of disclosure length and disclosure boilerplate. Our results suggest that lengthier and more boilerplate risk factor disclosures are less likely to be considered inadequate under judicial and regulatory review. Specifically, risk factors that are lengthier and less specific are less likely to be found inadequate by judges in shareholder securities lawsuits. In addition, more standardized risk factor disclosures are less likely to be targeted by an SEC comment letter during the SEC's filing review process. Further analysis finds that when risk factor language is assessed as adequate in judicial review, industry peers borrow that language more frequently. Finally, we find that judicial assessments of risk factor disclosures prompt industry peers to lengthen their own risk factor disclosures regardless of whether the risk disclosure was deemed adequate or not.

Keywords: risk disclosure; boilerplate; litigation risk; securities lawsuits; SEC comment letters; disclosure regulation

JEL classification: D8; G38; M4

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I. INTRODUCTION

The SEC mandates that firms disclose the most significant factors that make their stock speculative or risky in Item 1A of their periodic SEC filings (SEC 2005). These risk factor disclosures also play an important role in shareholder lawsuits because a safe harbor protects firms' forward-looking disclosures from litigation if they are accompanied by adequate cautionary language (15 U.S.C. § 78u5(c)). Regulatory and judicial guidance warns that firms should avoid lengthy and boilerplate risk disclosures in favor of disclosures that are concise and specific to the firm (SEC 1998; SEC 2005; SEC 2016; H.R. Conf. Rep. No. 104-369 (1995)). Recent research suggests that firms that deviate from this guidance by providing lengthy or boilerplate disclosures experience negative capital market consequences, such as higher cost of capital, greater stock price volatility, weaker market responses, and declines in analysts' ability to assess fundamental risk (e.g., Kravet and Muslu 2013; Campbell, Chen, Dhaliwal, Lu, and Steele 2014; Hope, Hu, and Lu 2016). However, practitioners and regulators lament that firms' risk factor disclosures continue to be generic and excessively long (e.g., Johnson 2010; IRRC 2016; SEC 2016; Berkman 2018). The fact that managers continue to produce lengthy and boilerplate disclosures that may expose them to negative capital market consequences suggests firms derive some offsetting benefit from this disclosure practice.

In this study, we propose and test the notion that lengthier and more boilerplate risk factor disclosures provide benefits to firms by reducing the likelihood those disclosures are flagged as inadequate under judicial and regulatory review. Risk factor disclosures are subject to both regulatory assessment under the SEC's filing review process and judicial assessment when federal judges consider whether a sued firm's forward-looking statements should be protected

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under the safe harbor.[1] Firms drafting Item 1A risk factor disclosures have incentives to minimize their total cost of disclosure, which include not only capital market consequences, but also costs arising from regulatory and judicial assessments of disclosure inadequacy. Thus, even if more concise and firm-specific risk factors provide greater clarity to investors, firms may believe lengthy and boilerplate risk factor disclosures help reduce their total costs of disclosure by decreasing the likelihood of unfavorable regulatory and judicial assessments.

Official guidance for regulators and the judiciary explicitly disfavors boilerplate disclosures and indicates risk factor disclosures should be concise and tailored to the specific risks faced by the firm (SEC 1998; SEC 2005; SEC 2016; Reg S-K 17 CFR 229.503; H.R. Conf. Rep. No. 104-369 (1995)).[2] Regulatory guidance indicates that boilerplate risk disclosures are disclosures that could apply to a broad set of issuers. Thus, boilerplate includes both disclosure language that is inherently non-specific (e.g., Hope et al. 2016) as well as standardized disclosure language that is so pervasively used among industry peers as to render it uninformative (e.g., Lang and Stice-Lawrence 2015). Although prior literature has considered both non-specificity and standardization to be proxies for disclosure "boilerplate," we focus on the two attributes separately to avoid the conflation of these two distinct disclosure practices.

Despite formal regulatory and judicial guidance, however, there are a number of reasons that lengthy, non-specific, and standardized risk factor disclosures may actually be associated

[1] Specifically, the Private Securities Litigation Reform Act (PSLRA) of 1995 provides a statutory safe harbor for firms' forward-looking statements accompanied by cautionary language describing the risks that could cause actual results to vary from projections.

[2] The terms "risk factor disclosures," "cautionary language," and "cautionary statement" are often used synonymously by legal scholars, courts, and legislators as either shorthand or the singular for "cautionary statements describing the risks that could cause actual results to vary from projections." (see e.g. Plumbers & Pipefitters Local Union No. 630 Pension-Annuity Trust Fund v. Allscripts-Misys Healthcare Solutions, Inc., 778 F. Supp. 2d 858 (2011)). We adopt a similar approach and refer to these disclosures generally as "risk factor disclosures," except when specifically referencing the safe harbor statute, in which case we will use the technical term "cautionary statements." Reg S-K 17 CFR 229.503 refers to the standard that was effective during our sample period prior to the May 2, 2019 modification.

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with more favorable judicial and regulatory assessments. First, although the PSLRA requires risk factor disclosures to convey "substantive" firm-specific information about risks, the PSLRA does not outline how "substantive" risk disclosures are identified. Published judicial opinions suggest that disclosure length is sometimes relied on as a heuristic for determining substantiveness.[3] Second, the legal doctrine of *stare decisis* (reliance on precedent) can lead judges to consider how closely a given set of risk factor disclosures for a sued firm resemble those already assessed as adequate in similar lawsuits. Third, lengthy and non-specific risk factor language may provide greater legal protection if projected results fail to materialize, as specific and concise risk factor disclosures inherently cover a narrower subset of adverse potential outcomes. Finally, the regulatory review process may actually favor standardized disclosure in spite of official guidance to the contrary, because the SEC's filing review process is conducted by staff with specialized industry expertise who review filings for a multitude of peers within the same industry.[4] Therefore, risk disclosures that are more similar to those of industry peers may be less likely to attract regulatory scrutiny.

We construct two separate samples to assess how risk factor disclosure length, non-specificity, and standardization correlate with the likelihood risk disclosures are assessed as inadequate under judicial or regulatory review. The first consists of judicial decisions ruling on firms' motion to dismiss securities lawsuits that allege false or misleading forward-looking statements. We focus on judges' explicit evaluation of firms' risk factor language in these judicial decisions because these evaluations provide the most direct indication of how risk factor

[3] For instance, in dismissing a suit against Motorola's allegedly false and misleading forward-looking statements, one judge reasoned that "the risk factors found in Motorola's SEC 10-K filings are extensive and specific, extending beyond eight pages." Silverman v. Motorola, Inc., 2008 U.S. Dist. LEXIS 76799, 2008 WL 4360648.
[4] See Division of Corporation Finance: Filing Review Process (SEC), available at https://www.sec.gov/divisions/corpfin/cffilingreview.htm (last updated June 27, 2018).

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attributes relate to safe harbor protections. To construct our second sample, we use Audit Analytics' SEC Comment Letter database to identify firms that received an SEC comment letter indicating a deficiency in the firm's risk factor disclosures.

To better understand the judicial and regulatory assessments and motivate our hypotheses, we also identify the specific reasons judges and regulators cite for why the risk factor disclosure was or was not adequate. We read all published judicial decisions relating to lawsuits alleging false or misleading forward-looking disclosures between 1996 to 2015. In these more than 500 decisions, we find that the judge explicitly assesses whether the firm's risk factor disclosures are adequate to provide safe harbor protection in 409 of these cases (over 80 percent of the time). Of these opinions, only 35 state that the risk disclosures were too vague, boilerplate, or non-specific to satisfy the safe harbor requirements, whereas 57 state that the risk factor was sufficiently specific or detailed. However, by far the most common rationale judges provide for why risk factor language is adequate is to simply state the language fulfils the statutory requirement to warn investors of risks that could cause actual results to vary. This qualitative evidence suggests that, despite common assertions that risk factor disclosures are excessively boilerplate and lengthy, judges generally find this language to be adequate for firms to obtain safe harbor protection. In contrast, we find SEC comment letters commonly ask firms to provide greater specificity or clarity in their risk factors. However, we also find the SEC is eight times as likely to request firms to add a risk factor than to remove one, suggesting that the SEC enforcement might not lead to more concise risk factor disclosure. Overall, we find significant variation in the rationale judges and the SEC offer to support their assessments of risk factor disclosure adequacy, suggesting this determination is far from straightforward.

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Our multivariate regression analyses indicate that the probability a federal judge rules that a firm's risk factor disclosures are inadequate for safe harbor protection is decreasing in disclosure length and specificity. These results are consistent with judges interpreting less specific language as covering a broader set of adverse potential outcomes and using length as a heuristic for adequate disclosure. Turning to the SEC comment letter sample, we find a strong negative association between our measure of standardized disclosure language and the probability of receiving an SEC comment letter related to risk factor disclosures. The finding that standardized risk disclosures are negatively associated with the likelihood of being flagged as inadequate by the SEC is consistent with the current structure of the SEC, which uses industry specialization of regional offices, leading SEC employees to develop a benchmark of adequate disclosure based on industry peers. These results are robust to controlling for determinants of firm risk, industry affiliation, fiscal year, the U.S. circuit in which the firm is headquartered, and the topics discussed in the firm's risk factor disclosures. Overall, our analyses indicate that more boilerplate disclosures are less likely to be flagged as inadequate under regulatory and judicial review, even after controlling for determinants of economic risk.

It is possible that some omitted factor drives both boilerplate risk factor language and favorable reviews of that language in our main analyses. To strengthen causal inferences of a link between adequacy assessments and disclosure, we next directly examine whether judicial assessments of sued firms' risk factor disclosure language impact the extent to which other non-sued firms adopt that same language in the future. Specifically, we use plagiarism detection software to measure the rate at which firms' risk factor language in any given year is borrowed by peers in subsequent years. We find that a firm's risk factor language becomes more heavily borrowed by peer firms in the years immediately following a judicial assessment that the firm's

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risk factor language is adequate for purposes of the safe harbor. We also find that firms increase the length of their risk factor disclosures after a judicial assessment of an industry peer's risk factor disclosures regardless of whether those disclosures are assessed favorably. These results corroborate inferences from our main analyses that generic, lengthy, and non-specific risk factor disclosures persist at least partly because firms believe adding risk factor disclosures that are not tailored to the specifics of the reporting firm can lead to more favorable judicial assessments of disclosure adequacy.

Our paper contributes to the stream of research examining the quality of firms' risk factor disclosures. Whereas recent research focuses on whether risk factor disclosures contain *any* information content, our study is among the first to propose and test an explanation for why lengthy, non-specific, and standardized risk factor disclosure continues to persist despite the potential for adverse capital market consequences. Our results suggest that such disclosure characteristics may benefit the firm by reducing expected legal and regulatory costs arising from judicial and regulatory enforcement. As a result, our study also extends prior research on the causes and consequences of the standardization of disclosure more generally (e.g., Kahan and Klausner 1997; McMullin 2016). Although McMullin (2016) suggests that the copying of peer firm disclosure is rampant, he does not explicitly identify factors that motivate this reporting behavior. Our analyses provide one explanation for the copying of and standardization of disclosures that are likely to come under regulatory or judicial review.

Our findings also contribute to three other streams of research. First, our results add to the literature on unintended consequences of disclosure regulation (e.g., Bushee and Leuz 2005; Arya, Glover, Mittendorf, and Narayanamoorthy 2005; Linck, Netter, and Yang 2009). Our study examines regulations that ostensibly promote concise and firm-specific risk disclosures.

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However, the evidence we find is consistent with the enforcement of these regulations rewarding lengthier, less specific, and more standardized risk factor disclosures. Second, our findings also contribute to the evidence regarding the efficacy of the PSLRA's safe harbor (e.g., Johnson, Kasznik, and Nelson 2001; Johnson, Nelson, and Pritchard 2007; Cazier, Merkley and Treu 2019) by explicitly considering the role of risk factor disclosures in judicial determinations of whether firms' forward-looking statements qualify for safe harbor protection. Finally, we also contribute to the text analysis literature by introducing a novel measure of disclosure borrowing.

Our findings have direct implications for regulators and policy makers seeking to understand the forces that shape current risk factor disclosure practices. The evidence we find suggests that one reason firms continue to provide lengthy, non-specific, and standardized risk factor language is that regulatory and legal outcomes generally do not reward disclosures that are concise, specific, and unique-to-the-firm. Our results also suggest that judges and the SEC may assess the adequacy of risk factor disclosures in a manner that is inconsistent with official guidance.

II. INSTITUTIONAL BACKGROUND AND HYPOTHESIS DEVELOPMENT

Capital Market Consequences of Risk Factor Disclosure Length and Boilerplate

Several recent studies have examined capital market outcomes associated with firms' risk factor disclosures. Kravet and Muslu (2013) find that increases in risk disclosure length is associated with subsequent increases in stock return volatility and trading volume and conclude that longer risk disclosures increase investors' perception of firm risk. Campbell et al. (2014) find that increases in risk factor disclosures increase firms' cost of capital, and suggest that this relation may provide strong incentives for managers to resist disclosure. Hope et al. (2016) find that more specific disclosures are associated with stronger market responses

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and with greater analyst ability to assess fundamental risk, and concurrent research finds that risk factor disclosures can predict firm-specific adverse events (Gaulin 2017). Overall, these studies suggest that risk factor disclosures that are less specific (i.e., more boilerplate) and lengthier can subject firms to negative capital market consequences.

One limitation of the settings examined in these prior studies is that financial statement users' assessment of risk factor disclosures is measured only indirectly via their response to the underlying 10-K filing, and hence financial statement users' response to the disclosure is potentially confounded with their response to the underlying economics. In contrast, we examine explicit judicial and regulatory assessments of the underlying risk factor disclosure *per se.* Our analysis of judicial and regulatory assessments of risk factor language is thus one step closer to shedding light on how characteristics of disclosure, rather than the economic events themselves, influence important corporate outcomes.

Although prior studies argue concise and specific risk disclosure are associated with positive capital market consequences, critics allege that risk factor disclosures continue to be excessively boilerplate and lengthy (e.g. Johnson 2010; IRRC 2016; SEC 2016; Berkman 2018). Consistent with critics' concerns about the informativeness of risk factor disclosures, Beatty, Cheng, and Zhang (2018) report that the market's response to unexpected 10-K risk factor disclosures has decreased in recent years. However, existing research sheds little light on the benefits firms derive from lengthy and boilerplate risk factor disclosures that could explain the persistence of these attributes. We contribute to this literature by examining how risk factor disclosure length, non-specificity, and standardization correlate with the likelihood regulators and federal judges will flag those risk factor disclosures as inadequate.

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As a preliminary matter, in order for judicial and regulatory outcomes to be a meaningful factor in firm disclosure behavior, firm management must be aware of these risks including judicial and regulatory outcomes. Our discussions with securities lawyers, while anecdotal, reveal that external securities lawyers are typically directly involved in drafting risk factor disclosure language on behalf of management. Consistent with lawyer involvement in risk factor disclosures, a practice advisor journal of a leading legal database provider provides practice tips and other guidance to lawyers for crafting risk factor disclosures (Gelfond, Wechsler, and Cohen 2019). In the case of SEC comment letters, prior research has shown that lawyers assist firms with the comment letter process (Bozanic, Choudhary and Merkley 2019) and that firms modify their risk factor disclosures when a peer firm receives an SEC comment letter (Brown, Tian and Tucker 2018), suggesting firms' Item 1A risk disclosures reflect an awareness of SEC comment letters sent to peers.

Risk Factors and the PSLRA's Safe Harbor

Risk factor disclosures play an important role in providing legal protections to firms making forward-looking statements. In response to concerns that securities lawsuits were discouraging managers from providing useful forward-looking information to investors, Congress enacted a safe harbor provision to protect forward-looking statements as part of the Private Securities Litigation Reform Act of 1995 (PSLRA) (15 U.S.C. § 78u5(c)).[5] In order to receive the full protection of the PSLRA safe harbor, firms must accompany forward-looking statements with "meaningful cautionary statements identifying important factors that could cause

[5] While the PSLRA codified a safe harbor protecting forward-looking statements, even prior to the PSLRA a common law rule called the Bespeaks Caution Doctrine protected projections and estimates accompanied by risk factors because misrepresented statements that were accompanied by such warnings were deemed to be immaterial. In re Trump Casino Sec. Litig., 7 F.3d 357, 371 (3d Cir. 1993); Harris v. IVAX Corp., 998 F. Supp. 1449, 1454 (S.D. Fla. 1998).

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actual results to differ." Unless an exception applies under 15 USC 78u-5(b), forward-looking statements are generally protected under the safe harbor when accompanied by meaningful cautionary statements (risk factors). This safe harbor protection generally results in the preliminarily dismissal of lawsuits brought by shareholders over allegedly false or misleading forward-looking statements.[6]

Congress did not explicitly define what makes cautionary language "meaningful", though the statute indicates that cautionary language must identify important risks that could cause actual results to vary from projections. The Congressional Conference Report accompanying the PSLRA asserts that risk factor disclosures should focus on specific risks, that boilerplate cautionary language would not suffice for purposes of the statute's safe harbor, and that only important factors should be listed (H.R. Conf. Rep. No. 104-369 (1995)).[7] After the SEC began mandating the disclosure of firms' principal risks as Item 1A in their periodic filings starting in 2005, Item 1A became the primary location for much of the cautionary language firms use to invoke safe harbor protection for their forward-looking statements (Nelson and Pritchard 2016).

Prior studies report evidence suggesting that the PSLRA's safe harbor is at least somewhat effective in protecting firms' forward-looking disclosures from legal liability (e.g.,

[6] The safe harbor of the PSLRA has two prongs and the first protects forward-looking statements that are immaterial and statements accompanied by "meaningful cautionary statements identifying important factors that could cause actual results to differ materially" (15 U.S.C. § 78u-5(c)(1)(A)). Claims based on statements determined to be covered by the first prong are typically dismissed by courts on preliminary motion without inquiry into the state of mind of the firm that made the statement. In the absence of such risk factors, lesser protections may apply under the second prong that permits an inquiry into the state of mind of the firm making the statement (15 U.S.C. § 78u-5(c)(1)(B)). The second prong protections are more similar to the general PSLRA protections for non-forward looking statements and so failure to achieve first prong protection can open the firm up to costly discovery proceedings.

[7] The Congressional Record is the official daily record of the proceedings and debates occurring in the United States Congress (See https://www.congress.gov/congressional-record.). While the congressional record does not carry the weight of law, the record provides insight into the purpose for the enactment of laws and courts frequently rely upon statements made in the congressional record when the interpretation of a particular statute is not clear. More importantly, this segment of the congressional record is highly cited by courts when analyzing the safe harbor.

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Johnson et al. 2001; Johnson et al. 2007; Cazier et al. 2019). However, we are aware of no research that examines the characteristics of risk factor disclosures that are more likely to provide safe harbor protections. Ours is the first study to explicitly consider the role of risk factor disclosures in judicial determinations of whether firms' forward-looking statements qualify for safe harbor protection. Ultimately, whether a firm's risk factor disclosures are deemed as adequate cautionary language depends on judicial interpretation and application of the statutory safe harbor provisions (e.g., Olazábal 2000; De Simone, Ingber, and Creutz 2004; Pritchard and Sale 2005; Cornerstone 2016). Thus, we focus on explicit judicial assessments of firms' risk factor disclosures to provide direct insight into how risk factor disclosure properties relate to safe harbor protections.

In order to gain institutional insight into the factors that judges claim influence their assessments of risk factor disclosure adequacy and to guide the development of our hypotheses, we read over five hundred judicial opinions from lawsuits filed between 1996 and 2015 relating to defendants' request to dismiss securities lawsuits that allege false or misleading forward-looking statements.[8] We find that in over eighty percent of these decisions, the judge makes an explicit ruling over the firms' risk factors as either adequate or inadequate to avail the firm's forward-looking statements of safe harbor protection. Next, we categorize the judges' rationale and tabulate in Table 1 the reasons cited by judges to support their rulings.

Panel A indicates that the extent to which risk factor language is boilerplate versus firm-specific is commonly cited as a factor influencing judicial assessments of risk factor disclosure adequacy. However, consistent with similar views expressed by legal scholars (Bloomenthal and

[8] Our initial search was performed using Lexis Advance Research and includes cases tagged by this service as securities cases with opinions issued between 1996 and 2017 that use the following phrases: "risk factor" or "meaningful cautionary language," and "safe harbor" or "forward-looking."

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Wolff 2012), our reading of the underlying judicial opinions suggests many of these assessments are largely ad hoc and are not based on any standard, objective benchmark of what constitutes sufficient specificity. For instance, Panel A of Table 1 indicates that several judges explicitly consider how similar a firm's risk factor disclosures are to risk disclosures that were considered adequate by prior courts. This practice is consistent with the legal doctrine of *stare decisis*, which binds judges to uphold precedents set in prior cases. In one case, a federal court granted a favorable ruling of risk factor disclosure adequacy because the firm's disclosure was "virtually identical to language approved by the Ninth Circuit in instances in which forward-looking statements were immunized by the PSLRA Safe Harbor."[9] Firms' adoption of risk factor language that has been vetted by prior courts could lead to more standardized risk factor language that is less likely to be considered inadequate under subsequent judicial review. Given these competing theories for the direction of the association between standardized risk factor disclosures and judicial assessments of inadequacy, we state our first hypothesis in null form:

H1a: There is no association between risk factor disclosure standardization and the likelihood risk factor disclosures will be deemed inadequate for purposes of the PSLRA's safe harbor.

The most common factor cited by judges ruling that risk factors are adequate is that the disclosure simply fulfils the requirement to warn investors of risks that could cause actual results to vary. The most commonly cited reason for why risk factor language is inadequate is the firm's omission of a material risk factor. Because specific risk factor disclosures inherently cover a narrower subset of potential adverse outcomes, it is possible that less specific risk factor

[9] In Re Fusion-IO, Inc. Securities litigation, 2015 U.S. Dist. LEXIS 18304; see also Plumbers & Pipefitters Local Union No. 630 Pension-Annuity Trust Fund v. Allscripts-Misys Healthcare Solutions, Inc., 778 F. Supp. 2d 858 (2011).

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disclosures provide superior legal protection. These considerations suggest that even generic risk factor disclosures may be adequate to avail firms of safe harbor protection. [10]

Consistent with this notion, one partner at a national law firm recently asserted that all 10-Ks contain boilerplate risk factor language to make such disclosure "as all-encompassing as possible" (see Berkman 2018). This incentive may help explain Hope et al.'s (2016) finding that Item 1A risk factor disclosures are less specific than other sections of the 10-K. Thus, despite legislative guidance that risk factors should be firm-specific, the actual association between risk factor specificity and judicial assessments of adequacy is unclear. We formalize the tension in the relationship between risk factor specificity and the likelihood risk factors are flagged as inadequate for purposes of the PSLRA's safe harbor as our next hypothesis, stated in null form as follows:

H1b: There is no association between risk factor disclosure specificity and the likelihood risk factor disclosures will be deemed inadequate for purposes of the PSLRA's safe harbor.

The safe harbor applies to any forward-looking statement, but only if that forward-looking statement is "accompanied by meaningful cautionary statements identifying important factors that could cause actual results to differ." 15 U.S.C. 78u-5(c)(A)(i). While courts must interpret what "important risk factors" means in the context of each case, federal regulations for risk factors in prospectus filings indicate that "significant risk factors" are "concise" and that registrants "should not present risk factors that could apply to any issuer or any offering" (17 CFR Sec. 229.503(c)). If courts apply reasoning similar to that found in these SEC regulations

[10] Bloomenthal and Wolff (2012) questioned whether the 7th circuit court correctly identified boilerplate language in determining the adequacy of risk factor disclosures in the case Asher v. Baxter International, Inc., 377 F.3d 727 (7th Cir. 2004) because such disclosures, although deemed adequate by the court, appeared to "cover as many conceivable generic factors that could relate to the company's business as possible."

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when determining the adequacy of risk factors for purposes of the safe harbor, then factors that are more concise may result in more favorable judicial outcomes.

However, there are a number of reasons that longer risk factor disclosures may provide superior legal protection. First, judges may view lengthier risk factor disclosures as better fulfilling the PSLRA's requirement that such disclosures convey "substantive" information about the risks that could cause actual results to vary. Panel A of Table 1 indicates that judges sometimes cite the length of a firm's risk factor disclosures as supporting evidence for their assertion that the firm's disclosures were adequate. Second, uncertainty regarding which risks are most important may lead assessors of risk disclosure adequacy to view lengthier risk factor disclosures as representing a good faith effort to warn of all potentially material risks the firm may face. Third, the inclusion of a longer list of risk factor disclosures helps ensure no obvious omissions have been made. Given that the congressional record and some evidence from our reading of judicial opinions may be at odds with one another, we formalize our next hypothesis in null form as follows:

H1c: There is no association between risk factor disclosure length and the likelihood risk factor disclosures will be deemed inadequate for purposes of the PSLRA's safe harbor.

The SEC's Item 1A Risk Factor Disclosure Mandate

In 2005, the SEC began mandating disclosure of firms' most significant risk factors in Item 1A of firms' periodic filings to help investors understand the nature of the risk inherent in the company. Prior to 2005, risk factor disclosures that firms provided for purposes of the PSLRA's safe harbor were not consistently reported in any particular section of firms' periodic filings. SEC reporting requirements stipulate that Item 1A risk factor disclosures should avoid

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vague boilerplate (SEC 1998; SEC 2012; 17 C.F.R. 230.421(b)) and be clear, concise, and tailored to the specific risks facing the reporting firm (e.g., SEC 2004; SEC 2005).

The SEC reviews Item 1A risk factor disclosures, along with firms' other periodic filings, as part of its filing review process. Firms whose risk factor disclosures are non-compliant with disclosure guidelines may receive SEC comment letters asking them to rectify the perceived deficiency. The SEC states it "concentrates its review resources on disclosures that appear to be inconsistent with Commission rules or applicable accounting standards, or that appear to be materially deficient in their rationale or clarity." The SEC Division of Corporate Finance's 11 offices possess industry-specific accounting and disclosure expertise and manage the review of filings of firms in these industries.[11]

The SEC's review of Item 1A risk factor disclosures varies in important respects from judicial review of risk factor disclosures to determine disclosure adequacy. Perhaps most importantly, the SEC is under no obligation to explicitly rule on the adequacy of risk factor disclosures that it reviews. Rather, the SEC only sends comment letters relating to risk factor disclosures if it identifies a perceived deficiency. Thus, no observable documentation exists regarding the SEC's assessments when risk factor disclosures are deemed to be adequate for regulatory compliance. To identify the specific disclosure deficiencies the SEC asks firms to rectify and to guide the development of our hypotheses, we read a randomly selected sample of two hundred SEC comment letters identified by Audit Analytics as relating to firms' risk factor disclosures. Based on our reading, we identified six separate categories of comments and tabulate them as Panel B in Table 1. Because the SEC may have multiple comments relating to risk factor disclosures within a single comment letter, the total number of comments tabulated in Panel B

[11] See https://www.sec.gov/divisions/corpfin/cffilingreview.htm (accessed on March 20, 2018).

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exceeds two hundred. Panel B indicates that the SEC often requests firms to clarify, expand, and occasionally reduce risk factor disclosures.

Although SEC comment letters often solicit additional information about the firm's own risks, it is possible that risk factor disclosures that more closely mirror industry peers' risks are less likely to attract regulatory scrutiny in the first place considering the structure of the SEC's process. The SEC's filing review process is conducted by staff who review filings for a large number of firms within the same industry, and as such the reviewing staff may be less likely to challenge risk disclosures that look similar to those of industry peers. Recent research suggests that firms preparing their own disclosures often look to peer firm disclosures to form their own expectation of what constitutes adequate disclosure, and that borrowing language from industry peers' disclosures is common (e.g., McMullin 2016, Berkman 2018; see also Brown et al. 2018). This process is likely facilitated by large law and audit firms' statutory reporting and disclosure assistance services, which provide financial reporting templates for their clients based on their knowledge of the types of disclosures that were flagged as inadequate by regulators in the past.[12]

Thus, despite the SEC's formal guidance that risk factor disclosures should be tailored to the firm, it is possible that more standardized risk factor language is less likely to attract regulatory attention. We formalize the tension in the relationship between standardized risk factor disclosures and the likelihood of an SEC comment letter as our next hypothesis, stated in null form as follows:

H2a: There is no association between risk factor disclosure standardization and the likelihood risk factor disclosures will be deemed inadequate under the SEC's filing review process.

[12] In addition, multiple subscription services exist that provide lawyers and auditors with updates on the outcomes of securities cases, including Lexis Nexis, Westlaw, and Bloomberg Law. More recently, databases have also been created that maintain outcomes from the SEC comment letter review process.

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Panel B of Table 1 indicates that a lack of specificity is the most frequently addressed issue in SEC comment letters relating to risk factor disclosures, suggesting that firms with more specific risk factor language may avoid receiving a comment letter from the SEC. However, a lack of disclosure specificity may not necessarily lead to a greater likelihood of receiving an SEC comment letter in the first place. The SEC's formal disclosure guidance requires firms to be concise while still disclosing the most significant risks that make their offering speculative or risky (SEC 2005). Pundits have argued that the SEC's dual requirements for both fulsome and concise disclosure lead to an inherent conflict in risk factor disclosure (Kuntz, Ehrenberg and Travis 2017). Because less specific risk factor language inherently covers a broader set of adverse potential outcomes concisely than does more specific language, disclosures that describe risks in less specific terms may be less likely to initially attract the SEC's attention. We consider the association between risk factor specificity and the likelihood risk factor disclosures are targeted by an SEC comment letter in our next hypothesis, stated in null form:

H2b: There is no association between risk factor disclosure specificity and the likelihood risk factor disclosures will be deemed inadequate under the SEC's filing review process.

Although SEC guidance cautions firms to make risk factors concise, lengthier risk factor disclosure may decrease the likelihood regulators will observe an obvious omission that could spawn further scrutiny. The descriptive evidence in Panel B of Table 1 suggests the SEC is more likely to request a firm to lengthen its risk factor disclosures rather than remove an existing factor. In 57 of the comment letters, the firm was asked to add an additional risk factor. In 40 comment letters the firm was asked to remove "qualifying language" and in only 7 cases did the SEC ask the firm to remove an existing risk factor. Thus, how risk factor disclosure length

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relates to the probability risk factor disclosures are targeted by the SEC is an open empirical question. We formulate our final hypotheses in null form as follows:

> **H2c: There is no association between risk factor disclosure length and the likelihood risk factor disclosures will be deemed inadequate under the SEC's filing review process.**

III. SAMPLE CONSTRUCTION AND VARIABLE MEASUREMENT

Sample Construction

Our analysis of judicial and regulatory assessments of firms' risk factor disclosures requires the construction of two distinct samples. The first sample consists of firms whose risk factor disclosures were assessed for adequacy by judges making safe harbor determinations during the dismissal phase of securities class action lawsuits. We search the Lexis Advance legal database for all published judicial opinions relating to the motion to dismiss securities litigation alleging false or misleading forward-looking statements between 2006 and 2017 (for lawsuits filed up to 2015). Keyword searches retrieved an initial judicial opinion sample for 328 securities lawsuits that we were able to match with lawsuit data from Stanford's Securities Class Action Clearinghouse database.[13]

We eliminate several cases due to missing Compustat or CRSP data and cases where either the judge determined the statements were not forward-looking or failed to explicitly rule on the adequacy of the firm's risk factor disclosures. Our procedure for selecting this sample is detailed in Panel A of Table 2. Ultimately, our regression analyses examining judicial assessments of risk factor disclosure are based on a sample of 144 distinct lawsuits in which the judge explicitly rules regarding the adequacy of the firm's risk factor language.

[13] Our initial search included cases between 1996 and 2017 and this search returned over 500 cases. This sample is reduced to 328 to align the time period with the presence of Item 1A in our sample period (2006-2015).

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We measure the judicial assessment of risk factor disclosure adequacy for each lawsuit and the boilerplate for the Item 1A Risk Factor disclosures in the firm's 10-K filed during the lawsuit's class period.[14] Because the class period of some lawsuits includes more than one year, our final sample for our main analysis of judicial assessments includes 246 unique firm-years corresponding to 246 unique 10-K filings made during the class period.

Our second sample consists of firm-years in which a firm received an SEC comment letter relating to at least one Item 1A risk factor disclosure between 2005 and 2015. We use Audit Analytics' SEC Comment Letter database to construct this sample. As illustrated in Panel B of Table 2, we begin with an initial sample of 4,450 firm-years for which Audit Analytics indicates the corresponding SEC filings were the target of a comment letter relating to risk factors. After eliminating firm-years for which Item 1A Risk Factor disclosures were not retrievable from the SEC website, firm-years for which textual analysis variables could not be computed from the extracted Item 1A disclosure, and firm-years missing necessary CRSP or Compustat data, we are left with a final sample of 1,666 firm-years for which risk factor disclosures were the target of an SEC comment letter in the following year.

We use this same dataset to identify firm-years in which a firm did not receive an SEC comment letter relating to risk factor disclosures to form a control sample. The SEC reviews filings of every publicly traded firm at least every three years, with some firms' filings being reviewed more frequently. However, the SEC does not reveal its formula for determining which firms to review more frequently, and in the absence of a comment letter, we cannot discern which firms were under review each year. We retain only firm-years in which the firm received

[14] In some cases, the class period did not include a 10-K filing date. In these cases, we selected the 10-K filed immediately prior to the start of the class period.

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at least one SEC comment letter during the year to ensure our sample includes only firms whose filings were actually under review. After identifying these firms, our final sample for the SEC comment letter analysis includes a total of 14,582 firm-years

Measuring Judicial and Regulatory Assessment of Risk Factor Inadequacy

Hypotheses 1a, 1b, and 1c center on the likelihood risk factor disclosures will be ruled inadequate under judicial review as a function of risk language standardization, specificity, and length. To code the dependent variable to test these hypotheses, we read each opinion in our sample of lawsuits and determine whether the judge assesses the firm's risk factor disclosures as adequate to avail itself of the PSLRA's safe harbor. We set the indicator variable *RF_INADEQUATE_JG* equal to one for firm-years in class periods in which the risk factor disclosures were ultimately deemed inadequate by the judge, and zero for firm-years in class periods for which the judge ruled the disclosures were adequate.

Hypotheses 2a, 2b, and 2c consider the probability risk factor disclosures will be flagged as inadequate under regulatory review as a function of risk language standardization, specificity, and length. We use the SEC's decision to issue a risk factor-related comment letter as a proxy for the SEC's assessment that the firm's risk factor disclosure was inadequate. We set the indicator variable *RF_INADEQUATE_SEC* equal to one if a firm-year's risk factor disclosures were targeted by an SEC comment letter, and zero otherwise. We use Audit Analytics' Comment Letter database to identify firms that received comment letters relating to their risk factor disclosures in the 10-K filing for any given year.

Our objective in testing our hypotheses is to capture the association between risk factor disclosure characteristics and judicial or regulatory assessments of risk factor disclosure adequacy, holding constant the firm's underlying risks, industry affiliation, headquarter location,

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and fiscal year. We next discuss our measure of disclosure standardization, specificity, and length, as well as the construction of each of our control variables.

Measures of Disclosure Length and Boilerplate

Prior literature has examined two measures of "boilerplate" that reflect distinct disclosure practices that cause disclosures to convey less firm-specific information. The first measure captures the extent to which disclosure language is so standardized that it is unlikely to be informative (e.g., Lang and Stice-Lawrence 2015). To capture firms' use of standardized language in their risk factor disclosure, we first identify commonly used trigrams (3-word phrases) in risk factor disclosure sentences for all firms in the same two-digit SIC industry. Next, we identify standardized sentences as those that either use 10 or more of these commonly used trigrams or for which 10 percent or more of the trigrams in the sentence are commonly used. We then count the number of words in these standardized sentences and divide this number by the total number of words in the risk factor section. Our approach to measuring this variable, which we label *STANDARDIZATION*, is similar to the approached used by Lang and Stice-Lawrence (2015) to measure boilerplate in annual reports of non-US firms.

The second measure of "boilerplate" is based on the Stanford Named Entity Recognition (NER) tool that determines the extent to which specific entities (person, organization, location, time, money, percent, and date) are discussed in the disclosure (e.g., Hope et al. 2016). This measure captures which disclosures are more or less specific based on the inherent precision of the language. Following Hope et al. (2016), we count the number of specific entities described in a firm's risk factor section and scale this word count by the total number of words in the risk factor section to adjust for differences in the length of this disclosure across firms. We multiply by negative one so higher values indicate less specific (i.e., more boilerplate) disclosure. We

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benchmark this measure against that of the firm's most similar industry peer. This benchmarking procedure accounts for the possibility that a firm's use of specific language may be a consequence of the underlying risks and economic activities of the firm, and we are interested in examining the effects of non-specific disclosure after controlling for the underlying determinants of the disclosure. Thus, this variable captures the extent to which firms' use of specific language in risk disclosures deviates from that of firms operating in similar economic environments. We identify a firm's most similar industry peer as the firm in the same two-digit SIC industry with the most similar 10-K Item 1 Business Description during the same time period. Appendix A contains greater detail on this matching procedure. We label this variable *NON_SPECIFICITY*.[15]

We measure Item 1A risk factor disclosure length based on the total number of words, consistent with how disclosure volume is measured by a host of prior studies (e.g., Li 2008; You and Zhang 2009; Miller 2010; Lawrence 2013; Campbell et al 2014). *LENGTH* is the natural logarithm of the number of words in the Item 1A risk factor disclosure in the firm-year's 10-K. Because we are interested in examining the association between risk factor disclosure length and judicial and regulatory assessments of adequacy after holding other determinants of disclosure constant, we also benchmark this variable against the most similar industry peer.

Control Variables

Our measures of risk factor disclosure length and boilerplate benchmark each firm's disclosure against those of peers, which helps control for the impact of underlying firm risks on adjudicators' assessments of disclosure adequacy. However, to account for any residual effects not controlled for in our matching procedure, we also include several measures of risk suggested

[15] Untabulated analyses indicate that NON_SPECIFICITY and STANDARDIZATION have less than a 20 percent correlation. Thus, although these two variables capture related constructs, the underlying disclosure behaviors are distinct.

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by prior literature. First, we include the correlation between each firm's stock returns with those of its industry peers to control for the fact that firms with similar underlying economic events may adopt similar language to describe those events. *RET_COR* is the mean correlation between each firm's 24-month return ending at the end of the current fiscal year and that of its 2-digit SIC industry peers.

We also construct several measures of firm risk that prior literature argues may influence firms' risk factor disclosures, and hence adjudicators' assessments of disclosure adequacy. *RET_VOL* is the standard deviation of daily returns for the year ending two days prior to the 10-K filing date. *BETA* is the firm's market beta measured using daily returns for the one year ending two trading days prior to the 10-K filing date. *SIZE* is the natural logarithm of the market value of equity. *LEV* is total liabilities scaled by total assets, measured as of the end of the fiscal year. *BIG_N* is an indicator variable equal to one if the firm is audited by one of the largest auditors (the Big N), and zero otherwise. *SKEW* is measured as the negative coefficient of skewness for daily returns over the year ending two days prior to the 10-K filing date. *TURNOVER* is the mean number of shares traded during the fiscal year scaled by the number of common shares outstanding. *ETR* is the firm's effective tax rate, computed as tax expense scaled by pre-tax income. *ROE* is income before extraordinary items scaled by the market value of equity. We include various fixed effects to control for industry affiliation, fiscal year, and the U.S. circuit in which each firm is headquartered because research indicates judicial practices and litigation risk sometimes vary by circuit (Hopkins 2018; Cazier, Christensen, Merkley, Treu 2018).[16] In some specifications we also include controls for the actual topics discussed in the

[16] We measure industry affiliation at the 2-digit SIC level for the SEC comment letter analysis. Because of the relatively small sample size in the lawsuit sample, we measure industry affiliation at the Fama-French 12 industry

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firm's risk factor disclosures (see Appendix A for construction of these risk topic controls), however, these controls are excluded from some specifications to address overfitting concerns.[17]

Equation 1 below shows the form of our primary regression model:

$$\Pr(Risk\ Factors\ Assessed\ as\ Inadequate_{i,t}) = \beta_0 + \beta_1 STANDARDIZED_{i,t} + \beta_2 NON_SPECIFICITY_{i,t} + \beta_3 LENGTH_{i,t} + \beta_4 RET_COR_{i,t} + \beta_5 RET_VOL_{i,t} + \beta_6 BETA_{i,t} + \beta_7 SIZE_{i,t} + \beta_8 LEV_{i,t} + \beta_9 BIG_N_{i,t} + \beta_{10} SKEWNEWSS_{i,t} + \beta_{11} TURNOVER_{i,t} + \beta_{12} ETR_{i,t} + \beta_{13} ROE_{i,t} + \text{Industry controls} + \text{Circuit controls} + \text{Year controls} + \text{Topic controls} + \varepsilon_{i,t} \quad (1)$$

Descriptive Statistics

Table 3 presents sample means for each of our independent variables of interest. Panel A displays the descriptive statistics for our sample of firms sued for false or misleading forward-looking statements. Panel B displays the descriptive statistics for firm-years in our comment letter analysis. We note that our variables do not appear to exhibit extreme skewness, as the mean for each continuous variable is within the inner-quartile range in nearly every case. To mitigate the effects of extreme outliers, all continuous variables are winsorized at the 1st and 99th percentiles.

IV. EMPIRICAL RESULTS

Judicial Assessment of Risk Factor Disclosure Adequacy

Our first set of hypotheses focuses on judicial assessments of risk factor disclosure inadequacy. To substantiate the notion that judicial assessments of risk factor disclosure inadequacy are associated with costly legal outcomes and to validate our measurement of *RF_INADEQUATE_JG*, we first estimate a logistic regression of the likelihood of lawsuit

classification level for those tests to maintain a practical number of degrees of freedom (http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/det_12_ind_port.html).

[17] Angrist and Pischke (2009, p. 225-226) note that while fixed effects effect estimates address a specific type of omitted variable, they may lead to attenuation bias. The analogy they offer is that including fixed effects "may kill some of the omitted variable bias bathwater, but they also remove much of the useful information in the baby, the variable of interest."

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dismissal as a function *RF_INADEQUATE_JG* and control variables. Column 1 of Table 4 displays results from estimating this regression over the entire sample period 1996 to 2015. Column 2 of Table 4 re-estimates this regression only on post-2005 observations for which we have Item 1A data. Both columns report a significantly negative coefficient on *RF_INADEQUATE_JG* ($p < 0.01$), confirming that a ruling of risk factor disclosure inadequacy for purposes of the safe harbor increases the likelihood the judge will dismiss the case. To test the sensitivity of our results to alternative specifications, we re-estimate the Table 4 regression under a linear model and report the results in Column 3 of Table 4. The coefficient estimate on *RF_INADEQUATE_JG* continues to be significantly negative in this specification as well.

Results from modeling judicial assessments of risk factor disclosure adequacy as a function of Item 1A standardization, non-specificity, and length individually and then collectively are reported in Table 5. The first column shows results of estimating a logistic regression model with industry, U.S. federal circuit, and year fixed effects included. The second column shows results from estimating the same linear probability model after including indicator variables controlling for the topics discussed in the firm's risk factor disclosure. We re-estimate these same two models under a linear probability model and report the results in columns 3 and 4 of Table 5. Standard errors are clustered by firm in all columns. Results indicate that *NON_SPECIFICITY* has a persistently negative coefficient estimate that is significant at the 5% level or higher in 3 out of the 4 models, and at the 10% level for the fourth. This negative coefficient estimate on *NON_SPECIFICITY* suggests risk factor disclosures that are less specific are *less* likely to be assessed as inadequate by judges for purposes of safe harbor protections.[18] The coefficient

[18] Stanford's NER tool identifies entities in seven categories, three of which refer to proper nouns (persons, organizations, and locations), and four of which refer to numeric values (time, money, percent, and date). In untabulated analyses, we split the *NON_SPECIFICITY* variable into the portion made of proper nouns versus

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estimate on *LENGTH* is negative and statistically significant at the 10% level or higher in all specifications, providing some evidence that longer risk disclosures are less likely to be deemed inadequate under judicial review. Finally, the coefficient estimate on *STANDARDIZATION* is negative but not statistically significant in any model of Table 5, providing no evidence that standardized language is associated with judicial assessments of risk disclosure adequacy.

Results in Table 5 suggest that, despite the caution expressed in the congressional record and by courts that risk factor disclosure should be specific and concise, less specific and lengthy risk factor disclosure correlates with more favorable judicial outcomes. These results are consistent with the view that lengthy and non-specific risk factor disclosure may provide greater legal protection by covering a broader set of potentially adverse events and, as occasionally expressed in judicial opinions, that relatively generic language can meet the requirements of the statute.[19]

To provide a sense of the economic magnitude of our results, we compute the mean marginal effects for the two explanatory variables of interest that load significantly in the last column of Table 5. Untabulated marginal effects suggest that going from the first quartile to the third quartile of *LENGTH* decreases the probability risk factor disclosures are ruled as inadequate under judicial review by 7.6 percent. The probability a firm's risk factor disclosures are ruled inadequate under judicial review decreases by 13.7 percent as those disclosures move from the

numeric values and find that the results in Table 5 appear to be driven primarily by a lack of specificity relating to proper nouns.

[19] An alternative interpretation of the results in Table 5 are that firms with fewer risks use less specific and more standardized language, and hence attract less legal scrutiny given their circumstances. We note, however, that *all* of the firms in our judicial analyses face a securities lawsuit, which often follows a precipitous drop in stock price, and hence all of these sample firms exhibited ex ante imminent risks that likely should have been disclosed in Item 1A. In addition, *all* sample firms in Table 4 faced legal scrutiny as the judge made an explicit evaluation regarding the adequacy of their risk factor disclosures. Results in Table 4 indicate that, conditional on being sued and facing legal scrutiny, firms' risk factor disclosures are more likely to be considered adequate when those disclosures are longer and less specific.

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first to the third quartile of *NON_SPECIFICITY*. This evidence suggests one reason firms may continue to issue lengthy and non-specific risk factor disclosures is to benefit from an increased likelihood of being granted safe harbor protection.

Regulatory Assessment of Risk Factor Disclosure Adequacy

Table 6 presents results from estimating equation (1) on our sample of firms for our SEC comment letter analysis. This analysis uses *RF_INADEQUATE_SEC* as the dependent variable which is equal to one for firm-years in which the firm receives an SEC comment letter relating to their Item 1A risk factor disclosures and zero otherwise. The ordering of results in Table 6 parallels Table 5. Estimating this model under both a logistic regression and a linear probability model results in a strongly negative coefficient estimate on *STANDARDIZATION*, significant at the 5% level or stronger in all specifications. This negative coefficient estimate suggests the SEC is significantly less likely to issue comment letters relating to risk factor disclosures when those disclosures use language that is pervasive in industry peers' disclosures. Neither *NON_SPECIFICITY* nor *LENGTH* are statistically correlated with the likelihood of receiving a risk factor-related SEC comment letter under any specification. Thus, despite SEC guidance indicating that only important risk factors should be included in firms' Item 1A disclosures, we observe no evidence that lengthier risk factors are disfavored in the SEC review process.

An analysis of marginal effects suggests that the association between standardized language (*STANDARDIZATION*) and the probability of an SEC comment letter targeting risk factor disclosures is relatively modest. Parameter estimates in Table 6 indicate that going from the first quartile to the third quartile of *STANDARDIZATION* decreases the probability of an SEC letter by less than 2 percent. However, results from Table 6 do suggest that, despite the SEC's frequent admonition to tailor risk factor disclosures to the specific risk factors faced by the firm,

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risk factor language that more closely resembles that of industry peers is less likely to be flagged as inadequate under regulatory review.[20]

Robustness Tests

To address the concern that our main results are attributable to differences in firm characteristics rather than differences in disclosure characteristics, we implement an alternative analysis technique to control for firm differences. Specifically, we construct high- and low-boilerplate and length subsamples and use entropy-balancing to reduce or eliminate covariate imbalance across each high and low sample pair. We estimate our main model using the entropy balancing weights after substituting the continuous measures of disclosure length, standardization, and non-specificity with an indicator denoting membership in the high text characteristic (treatment) subsample. Untabulated results support inferences from the main analysis. In addition, we find our primary results are robust to including a control variable that measures the cosine similarity between the risk factor topic probability vector of each firm's risk factors with those of its industry peers, and to substituting our measure of risk factor disclosure length with an alternative measure, proposed by Campbell et al (2014), which is based on the natural logarithm of a word count of specific risk-related words identified in their study.

We also examine the robustness of our main analyses to a variety of alternative cutoffs for identifying boilerplate trigrams used in measuring *STANDARDIZATION*. Our original construction of this variable defines boilerplate sentences as either: (1) those in which at least 10

[20] We conduct two untabulated robustness tests relating to Table 6. First, we add to the sample as additional control observations all firm-years in which the firm received *no* SEC comment letter. Second, we construct an alternative method of identifying comment letters relating to Item 1A risk factors, because we find that Audit Analytics' coding of an SEC comment letter as relating to "risk factors" often includes comments outside the scope of Item 1A. Specifically, we code an alternative dependent variable equal to one if the SEC comment letter includes the text "Item 1A," and zero otherwise. The results reported in Table 6 are qualitatively and quantitatively very similar to those produced by these alternative specifications.

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trigrams within the sentence are boilerplate, or (2) those in which 10 percent of all trigrams in the sentence are boilerplate. Boilerplate trigrams are defined as those appearing in at least 10 percent of all Item 1A disclosures in the same 2-digit SIC industry during our sample period, but not in more than 90 percent of all industry peers' Item 1A disclosures during that period. We find that our results are robust to changing these cutoffs to 20, 20, and 80 percent or 25, 25, and 75 percent, respectively. Finally, we examine results after broadening our measure of *STANDARDIZATION* to include trigram similarity with all firms in the economy rather than those in just the same 2-digit SIC industry. The resulting measure of standardized language arguably captures greater reliance on extremely generic risk factor disclosure language. We do not find any association between market-wide measures of standardized language based on any cutoffs and the likelihood of an SEC comment letter. Overall, our results suggest the SEC does not look either favorably or unfavorably upon risk factor language that is so broad that it could apply to *any* issuer.

V. BORROWING RATE AND LENGTH ANALYSIS

Our main analyses suggest that the incentive to avoid having risk factor disclosure flagged as "inadequate" under judicial or regulatory review may at least partially explain the prevalence of standardized, generic, and lengthy risk factor disclosures. We acknowledge results from our main analyses are only suggestive, as they focus on adjudicators' assessments rather than directly on firms' disclosure decisions. We next triangulate inferences from our main analyses by directly examining whether firms' disclosure behavior suggests they believe that adding language from risk factor disclosures recently vetted under judicial review influences the likelihood of their own favorable judicial and regulatory outcomes. Specifically, we examine whether firms' risk factor language becomes more likely to be borrowed by peer firms in the

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years following a judicial assessment in which those disclosures were deemed adequate for purposes of providing safe harbor protection.[21] We focus on the effects of judicial assessments in these analyses both because judicial assessments represent an economically more significant event than the receipt of an SEC comment letter and because SEC comment letters only indicate inadequate language, but not adequate language.

We start by developing a measure of the rate at which a firm's risk factor language becomes borrowed by other firms. Our empirical approach, explained in greater detail in Appendix A Section A.2, reflects the method firms commonly use to update their risk factor disclosures each year (e.g., Gelfond et al. 2019, Azarkh 2019, Kuntz et al 2017). This method consists of a firm reviewing the risk factor disclosures of several peer firms to identify any developing trends in risk disclosure or potential deficiencies in its own disclosure. The language identified during this review of other firms' disclosures becomes the starting point for what a firm adds to its own risk factor disclosures.

We use CopyFind, a plagiarism detection software program, to identify additions each firm makes to its Item 1A risk factor disclosure every year. We then compare these additions to other firms' risk factor disclosures filed in the preceding two years to determine whether these additions represent text borrowed from another firm's disclosure. This process, which involves over 11 billion distinct comparisons of text within CopyFind, produces a dataset of matched sequences of text strings of at least six words in length. We use these matched sequences of text strings to identify the set of firms whose previously filed disclosures contain text that most closely resembles the borrowed additions. We measure the rate at which a firm's risk factor

[21] By "borrowing" disclosure we mean a firm copies risk factor language of another firm and adds this language its own disclosure.

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disclosure is subsequently borrowed by other firms by counting the number of times in a given year that the firm is identified as a likely source firm for any other firm's risk factor disclosure additions containing borrowed text (*FREQ_SRCD)*. To shed light on the extent to which borrowing is concentrated among industry peers or firms sharing a common auditor, we also re-measure this rate based exclusively on borrowing from SIC 2-digit industry peers (*FREQ_SRCD_IND*), from peers audited by the same external auditor (*FREQ_SRCD_AUD*), and from firms that are neither in the same industry nor audited by the same external auditor (*FREQ_SRCD_NON*).

If a firm receives an adequate judicial assessment, we expect the judge's "stamp of approval" increases the rate at which other firms borrow this firm's disclosure. Finding evidence that adequate judicial assessments increase borrowing would support inferences from our main analyses that judicial outcomes contribute to the pervasiveness of risk factor boilerplate. Conversely, we expect that inadequate judicial assessments will have little effect on the rate at which a firm's disclosure is subsequently borrowed. To provide evidence on this issue, we estimate the following equation:

$$\text{FREQ_SRCD} = \beta_0 + \beta_1 \text{ADEQUATE}_{i,} + \beta_2 \text{ADEQUATE_POST}_{i\,i,} + \beta_3 \text{INADEQUATE}_{i,t} + \beta_4 \text{INADEQUATE_POST}_{i,t} + \text{Controls} + \text{Firm FE} + \text{Year FE} + \text{Topic FE} + \varepsilon_{i,t} \quad (2)$$

where *FREQ_SRCD* and its variants are calculated as described above. *ADEQUATE* is an indicator variable set equal to one in a two-year comparison window for firms receiving a judicial assessment that their risk factors were adequate, and zero otherwise. This comparison window includes the firm-year immediately prior to the lawsuit filing and the firm-year immediately after the judicial assessment. *ADEQUATE_POST* is an indicator variable set equal to one for the firm-year after the judicial assessment of adequacy, and zero otherwise. Hence, the coefficient estimate on *INADEQUATE_POST* measures the incremental rate at which firms' risk

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disclosures are borrowed by other firms immediately after the assessment of disclosure inadequacy relative to the year immediately prior to the lawsuit filing. *INADEQUATE* and *INADEQUATE_POST* are measured analogously for firms receiving an assessment of risk factor disclosure inadequacy. Controls include all control variables included in equation (1) as well as year and risk factor topic fixed effects. Because we have data for multiple firm-years for each firm in the sample, we are also able to include firm fixed effects in this model, though we note our inferences are unchanged if we replace the firm fixed effects with industry and circuit fixed effects as we do in the main analysis.

Because we compare each risk factor disclosure addition to other firms' risk factor disclosures made in the prior two years, our sample period for this analysis begins in 2007 to ensure we have sufficient years after the SEC's Item 1A mandate for firms to begin borrowing from peer firms' prior year disclosures. The sample period ends in 2015, which is the last full year for which we collected risk disclosures from EDGAR. We estimate this model for all firm-years for which we have CRSP, Compustat, and Item 1A risk factor data necessary for our model variables during our sample period. Thus, this sample primarily consists of non-sued firms that never had their risk factor disclosure assessed by a judge. Descriptive statistics for our regression variables in this analysis are provided in Table 7.

Table 8 presents results from estimating equation (2). We find strong evidence that judicial assessments indicating risk factor disclosure adequacy increase the extent to which a firm's risk factor disclosures are subsequently borrowed by other firms. The first column of Table 8 presents results from estimating equation (2) with *FREQ_SRCD* as the dependent variable. The resulting coefficient estimate on *ADEQUATE_POST* is positive and significant (0.296, p-value < 0.05), indicating that the frequency with which a firm's risk factor disclosures

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are borrowed increases in the year following a favorable judicial assessment. The coefficient estimate suggests borrowing increases 112 percent relative to the mean (0.296/0.265) following an adequate judicial assessment. To assess whether this increase in borrowing is concentrated among industry peer firms or firms that use the same external auditor as the sued firm, we estimate Models 2, 3, and 4. In these models, we set the dependent variable to *FREQ_SCRD_IND*, *FREQ_SRCD_AUD*, and *FREQ_SRCD_NON*, respectively. Table 8 indicates the coefficient on *ADEQUATE_POST* is nearly 3 times larger in Model 2 (0.299, p-value < 0.05) than in Model 3 (0.107, p-value > 0.10), suggesting the increase in borrowing is driven primarily by firms in the same industry, and not necessarily by firms that share the same auditor, as the sued firm. Model 4 indicates no evidence of an increase in borrowing among firms that are not connected to the sued firm via industry or external auditor. Finally, the coefficients on *INADEQUATE_POST* are not significantly different from zero across all four models, providing no support for the notion that an inadequate ruling influences the rate these sued firms' disclosures are borrowed.

Table 8 suggests a firm's risk factor disclosures are not more likely to be borrowed by peers after judicial assessments of disclosure inadequacy. However, we expect judicial assessments of inadequacy may still prompt industry peers to revise their risk factor disclosures in other ways. Specifically, peer firms may respond to a judge's assessment of risk factor disclosure inadequacy by adding new risk factor language to their own risk factors (possibly including language borrowed from non-sued peers) to minimize the risk their disclosures are assessed as inadequate in a future lawsuit. To test this conjecture, we re-estimate equation (2) after replacing the dependent variable in Table 8 with measures of risk factor disclosure length. We use the total number of words and the natural logarithm of the total number of words in the

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firm's Item 1A risk factor disclosure to measure risk factor disclosure length. For these analyses, we modify the *ADEQUATE* (*INADEQUATE*) variables to capture whether a 2-digit SIC industry *peer* firm's risk disclosure was assessed as adequate (inadequate) for safe harbor purposes. The interaction of these variables with *POST* allows us to compare the length of the firm's risk disclosure in the year after the judicial assessment relative to the year prior to the lawsuit filing. We present the results from this analysis in Table 9.

The coefficient estimate on *ADEQUATE_POST* in the first column of Table 9 is positive and statistically significant ($p < 0.05$). This result suggests firms' risk factor disclosures become longer after an industry peer's risk disclosures are assessed as adequate by a judge, and is consistent with evidence in Table 8 that firms adopt risk factor language from peers' disclosures that have been approved under judicial review. Interestingly, we find that the coefficient estimate on *INADEQUATE_POST* is also positive and strongly significant ($p < 0.01$). This coefficient estimate is over three times in magnitude the coefficient estimate on *ADEQUATE_POST*, suggesting firms add even more to their risk factor disclosures when peer disclosures are assessed as inadequate than they do when the peer disclosures are assessed as adequate. Thus, firms appear to behave as though they believe longer risk factors are more likely to lead to favorable judicial assessments of their risk factor disclosures.[22]

We perform various additional tests to examine the robustness of results reported in Tables 8 and 9. In untabulated analyses, we find inferences are unchanged after making the following adjustments: taking the natural log of the dependent variables in the borrowing

[22] For the sake of completeness, we also replicate our Table 8 and 9 analysis for our SEC comment letter sample. We find no consistent evidence that firms are less likely to borrow peers' risk factor disclosure language or more likely to increase the length of their own disclosures after an industry peer receives a risk factor-related SEC comment letter. This evidence is consistent with our priors that SEC comment letters represent a smaller economic event than do judicial outcomes.

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analysis or counting the number of sentences (instead of words) in the length analysis; estimating models with year, industry, or topic fixed effects alone; replacing firm and year fixed effects with industry-specific time trend variables; including topic fixed effects based on an LDA topic model with 20 topics instead of 200; and requiring sample firms to appear in both the pre-period and post period.[23] In addition, we estimate alternative specifications using entropy balancing to reduce any covariate imbalance across groups.[24] The entropy-balancing weighted regressions indicate consistent evidence of increased borrowing from sued firms that receive an adequate assessment and that non-sued peer firms increase the length of their risk factor disclosure in the year following the year an industry peer firm's risk language is judicially assessed.

Finally, we perform two placebo tests to ensure our results are not driven by a violation of the parallel trends assumption in our difference-in-differences tests. First, we identify the years prior to the pre-period as the pseudo-pre-period and the actual pre-period as the pseudo-post-period. Second, we identify the actual post period as the pseudo-pre-period and a year after the post-period as the pseudo-post-period. Both placebo tests indicate that our main results are not driven by an overall time trend in borrowing rates or risk factor disclosure length.

VI. CONCLUSION

Despite official reporting requirements from the SEC that Item 1A risk factor disclosures should be specific and tailored to the reporting firm, practitioners and regulators continue to

[23] We do not estimate the specification with industry specific time trends for the length analysis as the variables of interest in this analysis defined at the industry level.

[24] In the borrowing analysis, we identify weights first for observations in the sample of inadequate firms and then for observations in the non-sued control sample such that the weighted samples' covariate distributions (means, variances, and skews) are identical to those of the adequate firms. We also estimate two alternative specifications in the length analysis using entropy balancing. In the first alternative specification, we identify observation weights for firm-years in industries without an industry peer firm receiving a judicial review such that the weighted sample's covariate distributions (means, variances, and skew are identical to those firm-years where an industry peer received an adequate assessment. In the second alternative specification, we identify observation weights for the same control sample but balance to the sample of firm-years where an industry peer received an inadequate assessment.

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lament that these disclosures are boilerplate and uninformative. Whereas other academic studies suggest that risk factor disclosures convey *some* information to the market and that the use of lengthy boilerplate disclosures result in negative capital market consequences, our study is among the first to propose and test for an explanation for why firms continue to provide lengthy and boilerplate disclosures. Specifically, we provide evidence that lengthy, non-specific, and standardized risk factor disclosures are associated with more favorable regulatory and judicial outcomes.

Our supplemental analyses support inferences from our primary tests by showing that firms become more likely to borrow risk factor disclosures after those disclosures have been assessed as adequate under judicial or regulatory review. Thus, firms' disclosure behaviors suggest they *believe* adding borrowed risk factor language, which increases disclosure length and which is inherently not specific to the firm, decreases the likelihood their risk factor disclosures are assessed as inadequate under judicial review. In addition, firms' addition of risk factor language after peer firms' disclosures are assessed as inadequate suggests firms believe longer risk factor disclosures are more likely to lead to better judicial outcomes. Our results provide insight that may inform policy makers and standard setters regarding the potential unintended consequences of the ways current securities laws and regulations are written and enforced. Our results also highlight a potential disconnect between formal regulatory and legislative disclosure guidance and how that guidance is enforced via regulatory and judicial outcomes. Taken together, our results provide evidence on one explanation for the persistence of lengthy and boilerplate risk disclosures despite formal guidance indicating these disclosure characteristics are undesirable.

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APPENDIX A: Text Analysis of Risk Factor Disclosures

In this Appendix, we provide detailed explanations of our measures of risk factor disclosure length, standardization, and non-specificity, as well as our measure of the rate at which a firm's risk disclosure is subsequently borrowed by other firms.

SECTION A.1 - Measuring Risk Factor Disclosure Characteristics

Extracting Risk Factor Disclosures

To measure disclosure characteristics in Item 1A, we first use a PERL regular expression to extract Item 1A from form 10-K. This regular expression extraction is based on the headings which identify items (e.g., "Item 1A. Risk Factors", "Item 1B. Unresolved Staff Comments", for "Item 2. Description of Properties", etc.). The regular expression begins extracting text once it identifies the beginning of Item 1A and ceases extracting once it identifies the start of a subsequent item. To ensure the regular expression captures the heading of the item, we require the heading to be at the start of the line of text and that no other text on the same line follows the heading.

Description of *LENGTH*

To measure LENGTH, we count the number of words in Item 1A. Because the length of a firm's risk disclosure may be driven by the underlying risks and economic activities of the firm and since we are interested in the length of a firm's risk disclosure after controlling for the underlying determinants of the disclosure, we peer-adjust this count. Specifically, we take the natural log of this count and then subtract the natural log of the number of words in item 1A of a peer firm. We identify the peer firm as the firm with the most similar Item 1 Business Description during the same time period. This Item of the 10-K is where a company describes "its main products and services, what subsidiaries it owns,

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and what markets it operates in," and may include "information about recent events, competition the company faces, regulations that apply to it, labor issues, special operating costs, or seasonal factors."[25] This text serves as a good proxy for the main economic activity the firm undertakes (Hoberg and Phillips 2016). Specifically, we measure firm similarity between a firm and all other firms in the same two-digit SIC industry. Using the text of the 10-K's Item 1 Business Description, we estimate a topic model using Latent Dirichlet Allocation (LDA) with 200 topics, Blei, Ng, and Jordan (2003).[26] We then use this LDA model to calculate the topic probability vector (TPV) for each firm (Dyer, Lang, and Stice-Lawrence 2017). This topic probability vector contains the proportion of the Item 1 text that relates to each of the 200 topics and as a consequence the values sum to 100 percent. Using this TPV of 200 proportions, and in the same spirit as Hoberg and Phillips (2016), we calculate the cosine similarity score between two firms' TPVs. In using vectors indicating word usage in the text, Hoberg and Phillips (2016) assume that each word is a unique topic. Our use of the TPVs relaxes this assumption. We identify the most similar peer firm as the firm with the most similar TPV (highest cosine similarity). To peer-adjust *LENGTH*, we subtract the natural logarithm of the number of words in the peer's Item 1A.

Description of *NON_SPECIFICITY*

Using Python's natural language tool kit (NLTK) package (http://www.nltk.org), we process the text in firms' Item 1A through Stanford's Named Entity Recognition algorithm (Lafferty et al. 2001, Finkel et al. 2005). This algorithm identifies specific references to entities in seven categories: person, organization, location, money, date, time, and percent. Hope et al.

[25] See https://www.sec.gov/fast-answers/answersreada10khtm.html
[26] Blei (2012) provides an overview of topic models. Also, prior accounting studies have estimated topic models for 10-Ks (Dyer et al. 2017), analyst reports (Huang, Lehavy, Zang, and Zheng 2017), and risk factor disclosures (Campbell et al. 2014).

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(2016) explain in detail the implementation and assess the construct validity of this measure. We divide this count by the number of words in Item 1A and multiply by -1000 for exposition and so higher values capture non-specificity. Finally, we peer-adjust this measure in a similar manner to how we adjust *LENGTH*, as discussed above.

Description of *STANDARDIZATION*

Our approach to measuring this variable is similar to how Lang and Stice-Lawrence (2015) measure boilerplate in annual reports of non-US firms. Specifically, we measure standardized language in their risk factor disclosure by identifying commonly-used trigrams (3-word phrases) in risk factor disclosure sentences for all firms in the same two-digit SIC industry. We define commonly-used trigrams as trigrams that appear in at least 10 percent of all Item 1As, but no more than 90 percent of all Item 1As produced by firms in the same SIC 2-digit industry. Next, we identify standardized sentences as those that either use 10 or more of these commonly-used trigrams or for which 10 percent or more of the trigrams in the sentence are commonly used. After identifying standardized sentences, we then count the number of words in these standardized sentences and divide this number by the total number of words in the risk factor section. This *STANDARDIZATION* is the percentage of words in Item 1A that are in standardized sentences. We do not peer difference this measure as inherent in this measure is a comparison with industry peers' disclosures.

Description Risk Factor Disclosure Topic Measures

To capture the content discussed in the risk factor section, we estimate a topic model using a Bayesian machine-learning approach proposed by Blei et al. (2003). Specifically, we split each risk factor section into paragraphs, remove stop words, estimate an LDA topic model with 200 topics, and calculate a topic probability vector (TPV) for each risk factor paragraph. We

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identify the topic for each paragraph as the topic with the highest probability and create 200 indicator variables that equal one if the topic was disclosed in that firm's risk factor disclosures, and zero otherwise. We include these topic indicator variables in robustness tests to control for the content of the risk disclosure. Given the small sample size in our judicial outcomes analysis, to preserve degrees of freedom we estimate another LDA topic model with only 20 topics. Using this model we create 20 indicator variables and use these variables when controlling for topic inclusion in the lawsuit analysis.

SECTION A.2 - Measuring the Rate Firm *i*'s Risk Factor Disclosure is Borrowed

The purpose of the text analysis described below is to measure the rate text from one firm's Item 1A is subsequently borrowed by other firms. There are six steps in this measurement process. We describe these steps below and provide a figure illustrating this process.

Step 1 *– Identify new text added to risk factor disclosure in each firm-year.*

Using the plagiarism detection software, CopyFind, we compare Firm *j*'s year *t* Item 1A to Firm *j*'s year *t-1* Item 1A and extract text that is in Firm *j*'s Item 1A in Year *t* but not Firm *j*'s Item 1A in Year *t-1*.[27] These are the text additions (ADD_1 to ADD_K). Additions must be a least six words in length to be considered an addition. We select a six-word minimum because in a legal context even small phrases can have important implications, but selecting a lower minimum would inhibit our ability to reliably track the source of such new language. We identify 832,226 such additions in 46,178 10-Ks during our sample period. The mean risk factor section has 18 additions and the mean addition contains 58 words, as reported in Table A1 at the end of this appendix.

[27] We measure additions as text strings, at least six words in length, that are not in the firm's risk factor disclosure in the prior year. Lou Bloomfield, a physics professor at the University of Virginia, developed and makes freely available the CopyFind software. See https://plagiarism.bloomfieldmedia.com/software/copyfind/ to download the software and for details on the how the software detects similar text.

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Step 2 *– Determine if text additions contain borrowed language.*

Using CopyFind, we determine whether the text of each addition was borrowed from another firm's risk factor disclosure by comparing each addition to the Item 1As of other firms previously filed on EDGAR. Specifically, we compare each addition Firm *j* added to its risk factor disclosures (ADD_1 to ADD_K) in year *t* to all other (M) firms' Item 1As filed between January 1 of year *t-2* and the day prior to the filing of Firm *j*'s 10-K in year *t* (e.g., from 1/1/2010 to 2/10/2012 if Firm *j* filed its 10-K on 2/11/2012). Comparing each addition to each of the possible source Item 1As requires approximately 11 billion distinct comparisons.[28] CopyFind's Output Dataset reports $MATCHED_WORDS_{j,k,m,t-l}$ which is the number of words in text strings, at least six words in length, that appear in both borrowing Firm *j*'s ADD_k and potential source Firm *m*'s year *t-l* Item 1A document. We find that CopyFind reports no matches for 20.28% of the additions (N=168,810), suggesting that about one-fifth of all additions to risk factor disclosures represent original text. For the remaining 79.72% of the additions, CopyFind identifies text in the additions that also appears in one or more firms' previously filed risk factor disclosures indicating at least some of the new text is likely borrowed.

Step 3 *– Filter comparisons between Firm j's additions and other firms' Item 1As.*

CopyFind returns *any* evidence of borrowing, meaning it returns a match even if only a few words match across an addition and a possible source Item 1A. In this step, we filter this CopyFind dataset by keeping, for each addition, only possible source Item 1As where the number of matched words is the highest. This filter drops matches that likely do not identify the

[28] As the majority of borrowing likely is from the recent disclosures, we limit our comparisons to: (i) 10-Ks filed during the same calendar year, but prior to the file date of the 10-K containing the addition, as well as (ii) 10-Ks filed during the two full calendar years prior to the year the addition was filed. We use Indiana University's Karst supercomputer, a high-throughput computing cluster, to complete these comparisons. To allow for a sufficient source corpus, our analysis of these additions begins in 2007.

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primary source for the borrowed text. Specifically, we keep comparisons if $MATCHED_WORDS_{j,k,m,t-1}$ equals the maximum value of $MATCHED_WORDS_{j,k,m,t-1}$ for Firm *j*'s ADD_k. The point of this filter is to keep only those comparisons where the probability that Firm *j* borrowed from these firms' Item 1A is highest since they represent the comparisons with the greatest number of matching words. For each of Firm *j*'s additions, we count the number of possible source firms (not the number of matched Item 1As).

Step 4 – *Categorize additions into three categories.*

We are interested in identifying the borrowing of text that is not yet ubiquitous to examine how such text becomes standardized boilerplate and, as the number of potential source firms grows, our ability to plausibly identify the source of the borrowed text decreases. So we split all additions into three categories based on the number of their possible source firms. Category 1 includes additions where no borrowed text was found. Category 2 includes additions where at least 1 but no more than 24 possible source firms were found. Category 3 includes additions where more than 24 possible source firms were found. We retain only additions in Category 2 for the borrowing rate measurement as these additions include borrowed language that has not proliferated so broadly to make identifying the potential source unlikely as is the case with Category 3, which we exclude. We also exclude Category 1 from the borrowing rate analysis because these additions do not represent borrowing. After these exclusion filters, 390,371 risk factor additions remain and the mean (median) addition has 5.75 (3) possible source firms.

Admittedly, the source-firm cutoff of 24 is an arbitrary cut point that balances removing instances of a firm adding ubiquitous text while still allowing for other firms to have adopted the same text the firm is borrowing. For example, if we were to use a cutoff of one, then we would

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be missing cases where firm C borrows text from firm A *after* firm B borrows the same text from firm A since the plagiarism detection software would indicate two matches for firm C's addition; firm C to firm A and firm C to firm B. Steps 5 and 6 exploit the fact that firms often add multiple additions to their disclosure to focus in on the apparent source of borrowed text.

For descriptive purposes, we created indicators for whether an addition's text is likely borrowed from a firm in the same industry or that uses the same external auditor. Table A1 indicates that 53.1% of additions were likely sourced from a firm in the same Fama-French 12 industry (36.6% for 2-digit SIC, 21.3% for 4-digit SIC) and 40.4% of additions were likely sourced from a firm that shares the same auditor (see Table A1). This indicates that while industry and auditor connections influence borrowing, an extensive amount of borrowing occurs outside of industry or auditor affiliations. This also highlights the importance of searching for the source of borrowed text among all firms, rather than limiting the search to a firm's industry or among firms audited by the same auditor.

Step 5 *– Aggregate from the addition level to the Firm j level and identify probable source firms*

We next collapse the data from the addition-level to the Firm *j* Item 1A level. In doing this, we identify the set of firms from which a firm may have sourced the new language. For Firm *j*'s 10-K filed in year *t*, we identify the set of firms from which Firm *j* may have borrowed text when it inserted Category 2 additions into its Item 1A. We count the number of times these potential source firms appear as a possible source for any of Firm *j*'s Category 2 additions. We next create the set of firms that are the most likely source firms for Firm *j*'s borrowing in year *t* and label these firms the "Probable Source Firm Set." This set contains firms that were identified above as possible source firms for at least five additions in Firm *j*'s Item 1A disclosure in year

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t.[29] This last filter increases the amount of evidence required for a firm to be considered a *probable* source firm. Extending this point, although a firm may have borrowed from firms that do not fall into this refined set, these filters increase our confidence that our measure indeed captures borrowing and identifies firms from which a firm likely borrowed extensively.

Step 6 - *Measure the frequency with which Firm i's Item 1A disclosure is borrowed by other firms*

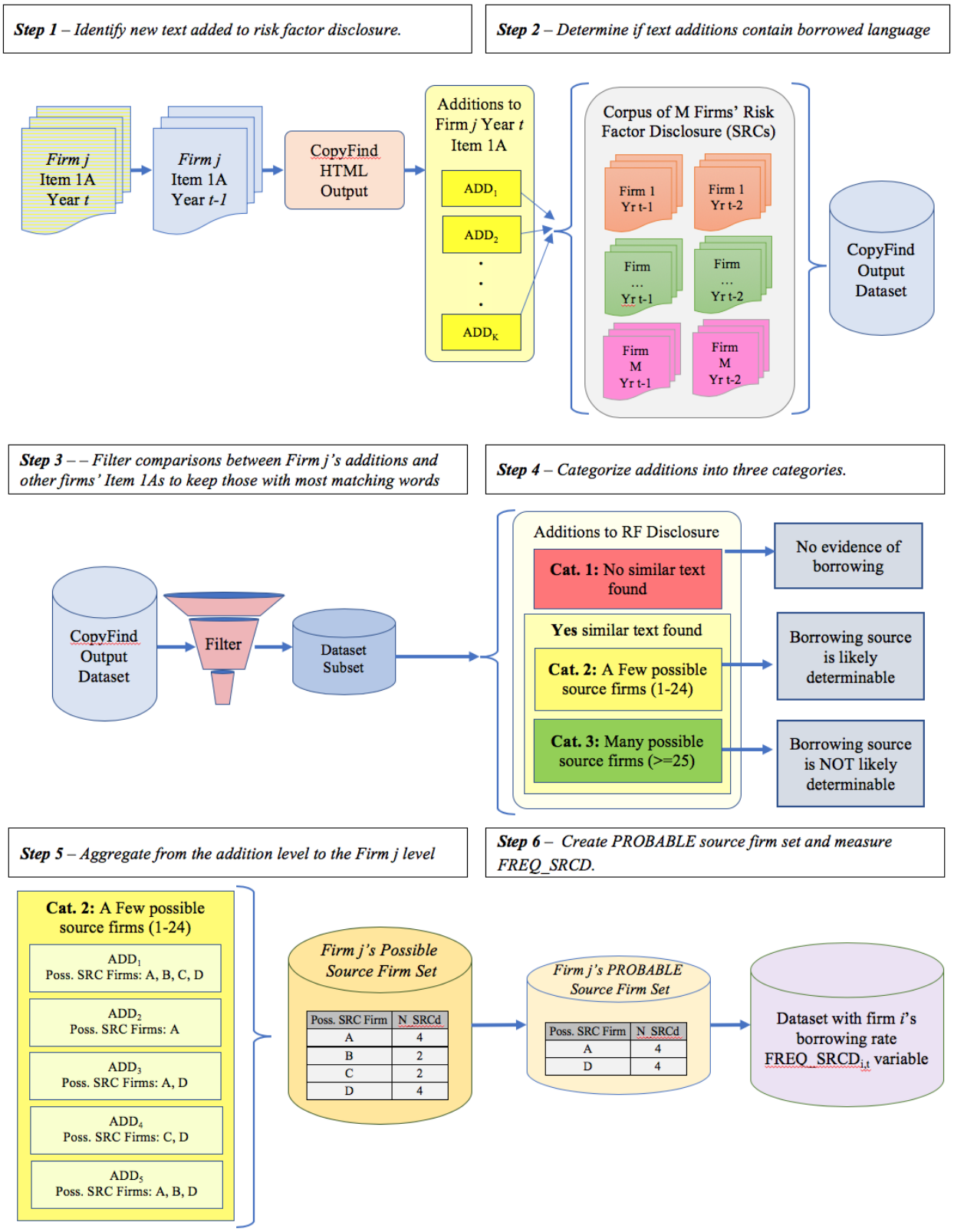
Finally, we construct the rate at which a firm's risk factor language is borrowed (*FREQ_SRCD*). Specifically, we count the number of times Firm *i* is included in any Firm *j*'s set of probable source firms during the year. Table 7 reports that the mean value of *FREQ_SRCD* is 0.254. *FREQ_SRCD_IND* is constructed by repeating the measurement process described above after dropping any potential source firm that is not in the same 2-digit industry as the borrowing firm. *FREQ_SRCD_AUD* is constructed by repeating this measurement process after dropping any potential source firms that are not audited by the same external auditor as the borrowing firm. Finally, we compute the borrowing rate from companies that are neither in the same industry nor audited by the same external auditor (*FREQ_SRCD_NON*). Table 7 reports the means for these variables as 0.108, 0.054, and 0.090, respectively.

The six steps in the process described above are illustrated in Figure A.1 below.

[29] Results are robust to using thresholds of either three or seven additions.

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Figure A.1 - Measuring the rate at which a Firm i's disclosure is borrowed



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TABLE A1

Descriptive Statistics of Text Additions to Item 1A

Panel A - Descriptive statistics of additions to firms' risk factor language

Firm-year sample of firms adding risk factor language (N = 46,178)

	Mean	St. Dev.	p25	Median	p75
Number of additions to Item 1A	18.023	20.489	5	12	24
Percent of additions containing any borrowed text	79.9%	21.2%	71.4%	83.9%	100.0%

All Additions to Risk Factor Disclosure (N=832,226)

	Mean	St. Dev.	p25	Median	p75
Number of words in the addition	57.981	99.665	11	21	56
MAX_MATCHED_WORDS divided by number of words in the addition	33.2%	31.8%	5.0%	23.7%	56.3%
Number of possible borrowing source firms	97.531	345.216	1	5	52
Indicator addition contains no borrowed text	0.203	0.402	0	0	0
Indicator addition contains borrowed text, less than 25 potential source firms	0.469	0.499	0	0	1
Indicator addition contains borrowed text, greater than 24 potential source firms	0.328	0.470	0	0	1

Additions with more than 1 and less than 25 potential source firms (N = 390,371)

	Mean	St. Dev.	p25	Median	p75
Number of words in the addition	54.865	106.521	12	20	43
MAX_MATCHED_WORDS$_{B,i,S,j}$ divided by Number of words in the addition	49.9%	28.2%	25.8%	45.0%	72.7%
Number of possible borrowing source firms	5.753	6.010	1	3	8
Indicator where at least 25% of addition's potential source firms are a(n):					
FF12 industry peer	0.531	0.499	0	1	1
SIC 2-digit industry peer	0.366	0.482	0	0	1
SIC 4-digit industry peer	0.213	0.409	0	0	0
Auditor peer (audited by the same auditor)	0.404	0.491	0	0	1

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APPENDIX B: Variable Definitions

Variable Name	Definition
NON_SPECIFICITY	The peer adjusted number of specific entities identified in the firm's risk factor section using the Stanford Named Entity Recognition tool, scaled by the total number of words in Item 1A and multiplied by -1,000. Additional details regarding the construction of this variable are available in Appendix A.
STANDARDIZATION	The number of words in boilerplate sentences in Item 1A, scaled by the total number of words in Item 1A. The identification of boilerplate sentences is detailed in Appendix A.
RET_COR	The mean correlation between the firm's 24-month return ending at the current fiscal year end and the same window return for each other firm in its same 2-digit industry.
RET_VOL	The standard deviation of daily returns for the year ending two days prior to the 10-K filing date.
BETA	The firm's market beta measured using daily returns for the one year ending two trading days prior to the 10-K filing date.
SIZE	The natural logarithm of the market value of equity, measured as of the end of the fiscal year.
LEV	Total liabilities scaled by total assets, measured as of the end of the fiscal year.
BIG_N	An indicator variable equal to one if the firm is audited by one of the largest (Big N) auditors, and zero otherwise.
SKEWNESS	The negative coefficient of skewness for daily returns over the year ending two days prior to the 10-K filing date.
TURNOVER	The mean number of shares traded during the fiscal year sacled by the number of common shares outstanding.
ROE	Income before extraordinary items, scaled by the market value of equity.
ETR	The firm's effective tax rate, computed as tax expense scaled by pre-tax income.
LENGTH	The natural log of the total number of words in the Item 1A risk factor disclosure. This variable is peer adjusted, where peers are identified as detailed in Appendix A.
RF_INADEQUATE_JG	An indicator variable equal to one for lawsuits in which the judge explicitly rules the firm's risk factor disclosures are inadequate to avail the firm of the PSLRA's safe harbor, and zero if the judge explicitly rules the risk disclosures are adequate.
RF_INADEQUATE_SEC	An indicator variable equal to one for firm-years for which the firm receives an SEC comment letter relating to risk factor disclosures in the 10-K for that fiscal year, and zero otherwise.

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APPENDIX B (continued)

Variable Name	Definition
$MATCHED_WORDS_{B,i,S,j}$	The number of words in text strings at least six words in length that appear in both borrowing firm B's addition i and source firm S's Item 1A j
$MAX_MATCHED_WORDS_{B,i}$	The maximum value of $MATCHED_WORDS_{B,i,S,j}$ for firm B's addition i
FREQ_SRCD	The number of times a firm-year's risk disclosure was identified as a probable source of other firm's borrowed additions of risk factor language.
FREQ_SRCD_IND	The number of times a firm-year's risk disclosure was identified as a probable source of other firm's borrowed additions of risk factor language. Borrowing firm must be in the same 2-digit SIC industry as the source firm.
FREQ_SRCD_AUD	The number of times a firm-year's risk disclosure was identified as a probable source of other firm's borrowed additions of risk factor language. Borrowing firm must be audited by the same external auditor as the source firm.
FREQ_SRCD_NON	The number of times a firm-year's risk disclosure was identified as a probable source of other firm's borrowed additions of risk factor language. Borrowing firm must neither be in the same 2-digit SIC industry nor audited by the same external auditor as the source firm.
INADEQUATE (ADEQUATE)	An indicator variable set equal to one for the year prior to and year after the year a firm receives an inadequate (adequate) assessment, zero otherwise. For the judicial assessment, the pre-period is based on the year the lawsuit is filed and the post-period is based on the year the judge rules on the motion to dismiss the lawsuit which is when the judge assesses the adequacy of the risk factor language. For the regulatory sample, the pre- and post-periods are based on the year the SEC sends a risk factor comment letter to the firm.
INADEQUATE_POST (ADEQUATE_POST)	An indicator variable set equal to one for the year after the year a firm receives an inadequate (adequate) assessment, zero otherwise, following the same convention for judicial and regulatory assessments as indicated in the description of INADEQUATE (ADEQUATE)
N_WORDS	The total number of words in the Item 1A risk factor disclosure.
LN_N_WORDS	The natural log of the total number of words in the Item 1A risk factor disclosure.

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APPENDIX B (continued)

Variable Name	Definition
INADEQUATE (IND. PEER) (ADEQUATE (IND. PEER))	An indicator variable set equal to one for the year prior to and year after the year a two-digit SIC industry peer firm receives an inadequate (adequate) assessment, zero otherwise, following the same convention for judicial and regulatory assessments as indicated in the description of INADEQUATE (ADEQUATE)
INADEQUATE_POST (IND. PEER) (ADEQUATE_POST (IND. PEER))	An indicator variable set equal to one for the year after the year a two-digit SIC industry peer firm receives an inadequate (adequate) assessment, zero otherwise, following the same convention for judicial and regulatory assessments as indicated in the description of INADEQUATE (ADEQUATE)

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TABLE 1

Factors Cited in Judicial and Regulatory Assessments of Risk Factor Disclosure Adequacy

Panel A: Judicial Assessments of Risk Factor Disclosure Adequacy for Safe Harbor Protections

Supporting Reasons Cited When Risk Factor Disclosure is Ruled Adequate		
Risk factor language fulfils the statutory requirement to warn investors of risks that could cause actual results to vary		94
Risk factor language is specific/detailed/tailored to the firm		57
Risk factor language included the risk that actually transpired		56
Assertion that risk factor language is adequate with no clear supporting explanation		47
Risk factor language is extensive/lengthy/numerous		30
Risk factor language is similar to that found to be adequate by prior courts		13
Other		12
	Total	309
Supporting Reasons Cited when Risk Factor Disclosure is Ruled Inadequate		
Failure to disclose a material risk factor		51
Failure to disclose important facts made risk factor disclosures misleading		46
Failure to provide any risk language with the forward-looking statements		35
Risk factor disclosure was too vague/boilerplate/non-specific		35
Adverse events disclosed as "risk factors" were actually happening		20
Other		12
	Total	199

Panel B: SEC Comments Relating to Risk Factor Disclosures

Increase specificity relating to information already disclosed		84
Explain further or clarify a risk factor		83
Add an additional risk factor		57
Remove qualifying language from risk factors		40
Change heading or formatting to make content of risk factor more clear		9
Remove a risk factor		7
	Total	280

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Panel A of Table 1 categorizes the factors cited by judges in explaining why a given firm's risk factor disclosures are deemed adequate or inadequate to provide legal protection to a defendant's forward-looking statements under the safe harbor of the PSLRA. Panel A is based on 409 unique judicial opinions regarding defendants' motion to dismiss lawsuits filed between 1996 and 2015. The total factors cited sum to more than 409 because in some cases the judge cites more than one unique factor. These lawsuits consist of 264 unique cases for which the risk factor disclosures were deemed adequate, and 145 unique cases for which the firm's risk factor disclosures were deemed inadequate. Panel B categorizes the SEC comments relating to firms' risk factor disclosures for a random sample of 200 risk factor related SEC comment letters issued between 2006 and 2015. The total issues cited in the SEC comment letters sums to more than 280 because in some comment letters the SEC requests more than one change to the risk factor disclosure.

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TABLE 2

Sample Selection

Panel A: Sample of Lawsuits Alleging False or Misleading Forward-Looking Statements

Securities lawsuits with published judicial opinions retrieved from keyword searches on Lexis Nexis Legal with a filing date in 2006 or later	328
Eliminate:	
Lawsuits against firms missing necessary Compustat or CRSP data or for which the CIK could not be found	(78)
Lawsuits that do not involve forward-looking statements	(37)
Lawsuits in which the judge does not explicitly rule regarding the adequacy of the firm's risk factors	(62)
Lawsuits missing Item 1A textual analysis data	(7)
Unique lawsuits examined in post-2005 multivariate regression analyses	**144**
Total 10-Ks filed during the class periods of the 144 lawsuits	**246**

Panel B: Sample of SEC Comment Letters Relating to Item 1A Risk Factor Disclosures

Firm-years receiving an SEC comment letter from 2005 to 2015 (from Audit Analytics)	35,931
Eliminate:	
Firm-years for which Item 1A textual analysis variables could not be computed	(17,266)
Firm-years missing CRSP and Compustat data variables necessary for model variables	(4,083)
Total firm-years included in SEC comment letter analysis	**14,582**

The final sample in Panel A consists of 151 (95) firm-year disclosures corresponding to lawsuits for which the underlying risk factor disclosure was considered adequate (inadequate). The final sample in Panel B consists of 1,666 SEC comment letters targeting risk factor disclosure, and 12,916 SEC comment letters not targeting risk factor disclosures.

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TABLE 3

Descriptive Statistics

Panel A: Descriptive Statistics for Firm-Years in Lawsuit Sample (N = 246)

Variable	Mean	Median	Q1	Q3	Std
LENGTH	0.0917	0.0915	-0.3757	0.4911	0.6612
NON_SPECIFICITY	-0.4687	-0.2724	-6.5086	5.8015	11.3285
STANDARDIZATION	0.5460	0.5518	0.4539	0.6421	0.1473
RET_COR	0.2151	0.2159	0.1247	0.2878	0.1222
RET_VOL	0.0343	0.0299	0.0223	0.0424	0.0179
BETA	1.2215	1.1775	0.9057	1.5446	0.4826
SIZE	7.4862	7.2264	6.3059	8.5297	1.8350
LEV	0.2299	0.1500	0.0028	0.3620	0.2478
BIG_N	0.8252	1.0000	1.0000	1.0000	0.3806
SKEW	0.3406	0.2648	-0.3176	0.9282	1.5483
TURNOVER	0.6031	0.4753	0.2665	0.8551	0.4609
ETR	0.1824	0.2888	0.0000	0.3702	0.3679
ROE	-0.0117	0.0430	-0.0326	0.0747	0.1903

Panel B: Descriptive Statistics for Firm-Years in SEC Comment Letter Sample (N = 14,582)

Variable	Mean	Median	Q1	Q3	Std
LENGTH	0.0013	0.0005	-0.4463	0.4431	0.7222
NON_SPECIFICITY	-0.4497	-0.1752	-6.3059	5.5801	11.1713
STANDARDIZATION	0.5462	0.5548	0.4673	0.6355	0.1349
RET_COR	0.2461	0.2385	0.1419	0.3461	0.1441
RET_VOL	0.0315	0.0266	0.0184	0.0392	0.0184
BETA	1.1096	1.0949	0.7618	1.4411	0.5314
SIZE	6.7180	6.7625	5.3337	8.0871	1.9810
LEV	0.2402	0.1860	0.0422	0.3667	0.2520
BIG_N	0.7675	1.0000	1.0000	1.0000	0.4225
SKEW	0.3688	0.2513	-0.1388	0.7452	1.2446
TURNOVER	0.5691	0.4012	0.2099	0.7146	0.5903
ETR	0.1757	0.2663	0.0000	0.3606	0.4220
ROE	-0.0252	0.0416	-0.0231	0.0721	0.2885

Variable definitions are provided in Appendix B. All continuous variables are winsorized at the 1st and 99th percentiles.

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TABLE 4

Inadequate Risk Factor Disclosures and the Probability of Lawsuit Dismissal

	Dependent Variable		
	DISMISSAL		
	Parameter Estimate *Standard Error*		
	Logistic Regression		*Linear Probability Model*
	Model 1	Model 2	Model 3
RF_INADEQUATE_JG	**-1.9931** ***	**-4.7041** ***	**-0.4320** ***
	0.2520	*1.0297*	*0.0605*
Risk control variables:			
RET_COR	**-0.3569**	**-0.1630**	**-0.0623**
	0.9688	*2.6787*	*0.2539*
RET_VOL	**3.0916**	**-34.0933**	**-1.7119**
	10.1762	*22.2919*	*2.0689*
BETA	**0.2138**	**0.9220**	**0.0361**
	0.2446	*0.7265*	*0.0612*
SIZE	**0.2687** ***	**0.0783**	**0.0125**
	0.0740	*0.1935*	*0.0182*
LEV	**-0.7639**	**-0.5730**	**-0.0385**
	0.5340	*1.5982*	*0.1143*
BIG_N	**-0.0386**	**0.1495**	**0.0271**
	0.4045	*0.9497*	*0.0789*
SKEW	**0.1535** *	**0.5262**	**0.0413** **
	0.0824	*0.1822*	*0.0164*
TURNOVER	**0.7542** ***	**1.9220**	**0.2120** ***
	0.2889	*0.7679*	*0.0613*
ETR	**0.3259**	**1.3903**	**0.0303**
	0.3159	*1.0701*	*0.0701*
ROE	**0.8553**	**0.3501**	**0.2032**
	0.7946	*2.8759*	*0.1775*
Industry fixed effects	Yes	Yes	Yes
Circuit fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Area under ROC curve	0.857	0.955	
R-Squared			0.595
N	524	237	237

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Table 4 displays the results of a logistic regression of the probability of a lawsuit dismissal on the indicator variable, RF_INADEQUATE_JG, and control variables. RF_INADEQUATE_JG is equal to one for lawsuits in which the judge explicitly rules the firm's risk factor disclosures are inadequate to avail the firm of the PSLRA's safe harbor, and zero if the judge explicitly rules the risk disclosures are adequate. All control variables are defined in the Appendix B. The sample includes all firm-years that have a 10-K filed within the class period of the lawsuit. The first column displays regression results for our entire sample of lawsuits from 1996 to 2015. The second column displays results for the post-2005 sample for which we have Item 1A textual data. *, **, and *** denote statistical significance at the .10, 0.05, and 0.01 significance levels, respectively (two-tailed tests).

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TABLE 5

Judicial Assessments of Risk Factor Disclosure Inadequacy

	Dependent Variable							
	RF_INADEQUATE_JG							
	Parameter Estimate *Standard Error*							
	Logistic Regression				*Linear Probability Model*			
	Model 1		Model 2		Model 3		Model 4	
STANDARDIZATION	**-1.9694**		**-1.8447**		**-0.3279**		**-0.1104**	
	2.2233		*3.5685*		*0.2919*		*0.3065*	
NON_SPECIFICITY	**-0.0547**	**	**-0.1527**	**	**-0.0058**	*	**-0.0084**	***
	0.0217		*0.0586*		*0.0031*		*0.0030*	
LENGTH	**-1.1369**	**	**-1.2118**	*	**-0.1311**	**	**-0.1288**	***
	0.4344		*0.7298*		*0.0547*		*0.0489*	
Risk control variables:								
RET_COR	**-0.0851**		**1.5849**		**-0.1017**		**-0.1399**	
	2.4124		*4.6377*		*0.3163*		*0.2891*	
RET_VOL	**17.1857**		**13.3144**		**2.6341**		**2.9843**	
	22.7299		*60.8282*		*2.3668*		*2.6614*	
BETA	**-0.3332**		**-1.8279**	*	**-0.0409**		**-0.0852**	
	0.5667		*1.0203*		*0.0863*		*0.0827*	
SIZE	**-0.0238**		**-0.2193**		**-0.0094**		**-0.0092**	
	0.2010		*0.4025*		*0.0279*		*0.0280*	
LEVERAGE	**2.5037**	**	**2.4886**		**0.3416**		**0.1221**	
	1.0808		*2.2107*		*0.1635*		*0.1719*	
BIG_N_AUDITOR	**-0.4815**		**-2.5447**	*	**-0.0880**		**-0.1240**	
	0.8594		*1.4052*		*0.1152*		*0.1138*	
SKEWNESS	**0.2147**	*	**0.1341**		**0.0304**		**0.0122**	
	0.1329		*0.3530*		*0.0142*		*0.0159*	
TURNOVER	**0.0393**		**0.0292**		**0.0024**		**0.0008**	
	0.4578		*0.8855*		*0.0774*		*0.0712*	
ETR	**-0.0811**		**0.6334**		**0.0297**		**0.0744**	
	0.7543		*1.0163*		*0.0838*		*0.0833*	
ROE	**6.5395**	**	**9.5208**	**	**0.7837**		**0.6622**	***
	1.8375		*4.5805*		*0.2170*		*0.2090*	
Industry fixed effects	Yes		Yes		Yes		Yes	
Circuit fixed effects	Yes		Yes		Yes		Yes	
Year fixed effects	Yes		Yes		Yes		Yes	
Topic fixed effects	No		Yes		No		Yes	
R-squared					0.430		0.574	
Area under ROC curve	0.900		0.967					
N	246		246		246		246	

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Table 5 reports the results from a regression of the probability the sued firm's risk factor disclosures are assessed as inadequate by a judge for purposes safe harbor protection under the PSLRA. *RF_INADEQUATE_JG* is equal to one if the firm's risk disclosure language is explicitly ruled as inadequate for purposes of the PSLRA's safe harbor, and zero otherwise. These assessments are based on the published judicial opinion supporting the judicial decision whether to grant the defendant's motion to dismiss the securities lawsuit. Models 1 and 2 estimate a logistic regression model. Models 3 and 4 estimate the same model using linear regression. Other variable definitions are provided in Appendix B. Industry fixed effects are measured at the Fama/French 12-industry classification level. Topic indicators are included based on twenty topics estimated using Latent Dirichlet Allocation over the entire corpus of 10-K Item 1A risk factor disclosures over our sample period. All continuous variables are winsorized at the 1st and 99th percentiles. Heteroskedasticity-consistent standard errors are used to assess statistical significance, which is denoted by *, **, and *** for statistical significance at the 0.10, 0.05, and 0.01 levels, respectively.

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TABLE 6

Regulatory Assessments of Risk Factor Disclosure Inadequacy

	Dependent Variable							
	RF_INADEQUATE_SEC							
	Parameter Estimate *Standard Error*							
	Logistic Regression				*Linear Probability Model*			
	Model 1		Model 2		Model 3		Model 4	
STANDARDIZATION	**-0.9546**	***	**-0.7579**	**	**-0.0952**	***	**-0.0782**	***
	0.256		*0.2974*		*0.0267*		*0.0296*	
NON_SPECIFICITY	**-0.0016**		**-0.0036**		**-0.0002**		**-0.0004**	
	0.0025		*0.0027*		*0.0003*		*0.0003*	
LENGTH	**-0.0125**		**-0.0633**		**-0.0013**		**-0.0055**	
	0.0393		*0.0444*		*0.0039*		*0.0043*	
Risk control variables:								
RET_COR	**-0.1544**		**-0.0627**		**-0.0175**		**-0.0092**	
	0.275		*0.2895*		*0.0278*		*0.0284*	
RET_VOL	**10.6496**	***	**8.7335**	***	**1.2721**	***	**1.0626**	
	2.5953		*2.8035*		*0.2971*		*0.3103*	
BETA	**0.0123**		**0.0003**		**0.0015**		**0.0011**	
	0.0633		*0.0665*		*0.0064*		*0.0065*	
SIZE	**0.0564**	**	**0.0538**	**	**0.0066**	***	**0.0066**	
	0.0218		*0.0245*		*0.0021*		*0.0024*	
LEV	**-0.1757**		**-0.2024**		**-0.0165**		**-0.0175**	
	0.1202		*0.1385*		*0.0107*		*0.0116*	
BIG_N	**-0.1826**	**	**-0.1138**		**-0.0180**	**	**-0.0123**	
	0.0753		*0.0801*		*0.0077*		*0.0078*	
SKEW	**0.0154**		**0.0209**		**0.0016**		**0.0021**	
	0.0218		*0.0232*		*0.0020*		*0.0020*	
TURNOVER	**0.0001**		**-0.0025**		**-0.0010**		**-0.0008**	
	0.0511		*0.0553*		*0.0048*		*0.0051*	
ETR	**-0.0281**		**-0.0389**		**-0.0016**		**-0.0031**	
	0.0695		*0.0716*		*0.0068*		*0.0069*	
ROE	**-0.0447**		**-0.0473**		**-0.0082**		**-0.0100**	
	0.0934		*0.0946*		*0.0122*		*0.0125*	
Industry fixed effects	Yes		Yes		Yes		Yes	
Circuit fixed effects	Yes		Yes		Yes		Yes	
Year fixed effects	Yes		Yes		Yes		Yes	
Topic fixed effects	No		Yes		No		Yes	
R-squared					0.044		0.062	
Area under ROC curve	0.688		0.719					
N	14,582		14,582		14,582		14,582	

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Table 6 reports the results from estimating a regression of the probability that a firm's risk factor disclosures are targeted by an SEC comment letter during the SEC's filing review process. Models 1 and 2 estimate a logistic regression model. Models 3 and 4 estimate the same model using linear regression. Variable definitions are provided in Appendix B. Industry fixed effects are measured at the two-digit SIC classification level. Topic indicators are included based on two hundred topics estimated using Latent Dirichlet Allocation over the entire corpus of 10-K Item 1A risk factor disclosures over our sample period. All continuous variables are winsorized at the 1st and 99th percentiles. Robust standard errors are used to assess statistical significance, which is denoted by *, **, and *** for statistical significance at the 0.10, 0.05, and 0.01 levels, respectively.

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TABLE 7

Descriptive Statistics for Firm-Years in Borrowing and Length Sample

Full Sample (N = 24,756)

	Mean	St. Dev.	p25	Median	p75
Borrowing Analysis Variables					
FREQ_SRCD	0.265	0.742	0	0	0
FREQ_SRCD_IND	0.110	0.473	0	0	0
FREQ_SRCD_AUD	0.056	0.308	0	0	0
FREQ_SRCD_NON	0.097	0.370	0	0	0
ADEQUATE	0.005	0.067	0	0	0
INADEQUATE	0.002	0.047	0	0	0
Length Analysis Variables					
N_WORDS	6,718.6	4,665.3	3,520.5	5,684.5	8,704.0
LOG_N_WORDS	8.593	0.691	8.166	8.645	9.072
ADEQUATE (IND. PEER)	0.438	0.496	0	0	1
INADEQUATE (IND. PEER)	0.252	0.434	0	0	1
Control Variables					
RET_COR	0.245	0.146	0.140	0.239	0.344
RET_VOL	0.032	0.019	0.019	0.027	0.039
BETA	1.076	0.558	0.710	1.080	1.420
SIZE	6.399	1.970	5.026	6.373	7.722
LEVERAGE	0.215	0.236	0.023	0.156	0.329
BIG_N_AUDITOR	0.728	0.445	0	1	1
SKEWNESS	0.391	1.390	-0.126	0.258	0.734
TURNOVER	0.587	1.089	0.195	0.389	0.695
ETR	0.181	0.444	0.005	0.275	0.363
ROE	-0.013	0.218	-0.019	0.044	0.075

Table 7 provides descriptive statistics of the variables in the multivariate regressions in Tables 8 and 9. Variable definitions are provided in Appendix B.

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TABLE 8

Borrowing Analysis of Firms with Judicially Assessed Risk Factor Disclosure

	Parameter Estimate *Standard Error*			
	Model 1	Model 2	Model 3	Model 4
Dependent Variable:	FREQ_SRCD	FREQ_SRCD_IND	FREQ_SRCD_AUD	FREQ_SRCD_NON
ADEQUATE	**-0.157**	**-0.134****	**-0.016**	**0.014**
	0.096	*0.067*	*0.021*	*0.057*
ADEQUATE_POST	**0.296****	**0.299****	**0.107**	**-0.085**
	0.138	*0.119*	*0.072*	*0.078*
INADEQUATE	**0.210**	**0.172**	**0.168**	**0.014**
	0.198	*0.109*	*0.111*	*0.091*
INADEQUATE_POST	**-0.058**	**-0.077**	**-0.126**	**0.001**
	0.198	*0.097*	*0.149*	*0.118*
RET_COR	**0.067**	**0.053**	**0.013**	**-0.049***
	0.048	*0.033*	*0.021*	*0.028*
RET_VOL	**0.850**	**-0.221**	**-0.224**	**0.231**
	0.539	*0.324*	*0.191*	*0.280*
BETA	**-0.009**	**0.003**	**-0.009**	**-0.015***
	0.014	*0.010*	*0.006*	*0.008*
SIZE	**0.025****	**0.010**	**0.005**	**0.023*****
	0.012	*0.008*	*0.005*	*0.006*
LEVERAGE	**0.020***	**0.000**	**0.003**	**0.003**
	0.010	*0.008*	*0.004*	*0.004*
BIG_N_AUDITOR	**0.008**	**-0.005**	**0.002**	**0.006**
	0.024	*0.016*	*0.010*	*0.014*
SKEW	**-0.009*****	**-0.005****	**-0.003****	**-0.002**
	0.003	*0.002*	*0.001*	*0.002*
TURNOVER	**0.004**	**0.004**	**0.001**	**0.001**
	0.007	*0.003*	*0.002*	*0.006*
ETR	**0.001**	**0.001**	**0.005**	**-0.006**
	0.009	*0.005*	*0.004*	*0.005*
ROE	**0.000**	**-0.021**	**-0.017**	**-0.000**
	0.026	*0.018*	*0.010*	*0.014*
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Topic FE	Yes	Yes	Yes	Yes
Observations	24,756	24,756	24,756	24,756
Adjusted R^2	0.371	0.376	0.282	0.141

Table 8 reports the results from estimating a regression of the rate firms' Item 1A disclosures are borrowed by peer firms. Adequacy of risk factor disclosure is assessed by judges when the firms are sued and file a motion to dismiss. Variable definitions are provided in Appendix B. Industry fixed effects are measured at the SIC 2-digit industry classification level. Topic indicators are included based on two hundred topics estimated using Latent Dirichlet Allocation over the entire corpus of 10-K Item 1A risk factor disclosures over our sample period. Standard errors are clustered by firm. All continuous variables are winsorized at the 1st and 99th percentiles. Statistical significance is denoted by *, **, and *** significance at the 0.10, 0.05, and 0.01 levels, respectively.

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TABLE 9

Analysis of Firms' Risk Factor Disclosure Length When Industry Peers' Risk Factor Disclosure are Judicially Assessed

	Parameter Estimate *Standard Error*	
	Model 1	Model 2
Dependent Variable:	N_WORDS	LN_N_WORDS
ADEQUATE (IND. PEER)	**-16.281**	**-0.025*** **
	37.556	*0.005*
ADEQUATE_POST (IND. PEER)	**92.398** **	**0.021*** **
	37.647	*0.004*
INADEQUATE (IND. PEER)	**-140.521*** **	**-0.014*** **
	34.530	*0.005*
INADEQUATE_POST (IND. PEER)	**301.857*** **	**0.032*** **
	59.339	*0.006*
RET_COR	**364.782*** **	**0.068*** **
	137.800	*0.019*
RET_VOL	**7927.960*** **	**1.386*** **
	1840.969	*0.198*
BETA	**14.673**	**-0.001**
	40.709	*0.005*
SIZE	**167.941*** **	**0.011** **
	40.598	*0.005*
LEVERAGE	**407.655***	**0.058** **
	215.262	*0.027*
BIG_N_AUDITOR	**120.067**	**0.012**
	100.385	*0.015*
SKEW	**-20.314***	**-0.004*** **
	12.054	*0.001*
TURNOVER	**-6.685**	**0.001**
	24.384	*0.003*
ETR	**53.819** **	**0.003**
	27.337	*0.003*
ROE	**-302.200*** **	**-0.043*** **
	96.406	*0.010*
Firm FE	Yes	Yes
Year FE	Yes	Yes
Topic FE	Yes	Yes
Observations	24,756	24,756
Adjusted R^2	0.890	0.919

Table 9 reports the results from estimating a regression of the length of firms' Item 1A disclosure. Inadequacy of risk factor disclosure is assessed by judges when the firms are sued and file a motion to dismiss. Indicators capture whether a peer firm in the same 2-digit industry had its risk factor language assessed during a lawsuit. Variable definitions are provided in Appendix B. Industry fixed effects are measured at the SIC 2-digit industry classification level. Topic indicators are included based on two hundred topics estimated using Latent Dirichlet Allocation over the entire corpus of 10-K Item 1A risk factor disclosures over our sample period. Standard errors are clustered by firm. All continuous variables are winsorized at the 1st

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and 99th percentiles. Statistical significance is denoted by *, **, and *** significance at the 0.10, 0.05, and 0.01 levels, respectively.

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